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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Structure of Atom

JEE Main Chemistry · Previous Year Questions with Answer Key

233	190	2002 – 2026	4	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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Numerical	56 questions
MCQ (Single Correct Answer)	173 questions
Out of syllabus	4 questions
Answer key	at the end

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Numerical 56 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Consider two radiations of wavelengths

1. $\lambda_1 = 2000\overset{o}{A}$
2. $\lambda_2 = 6000\overset{o}{A}$

The ratio of the energies of these two radiations $\left(\frac{E_1}{E_2}\right)$ is ____ (Nearest integer).

Open solution

Q2. JEE Main 2026 (Online) 2nd April Evening Shift

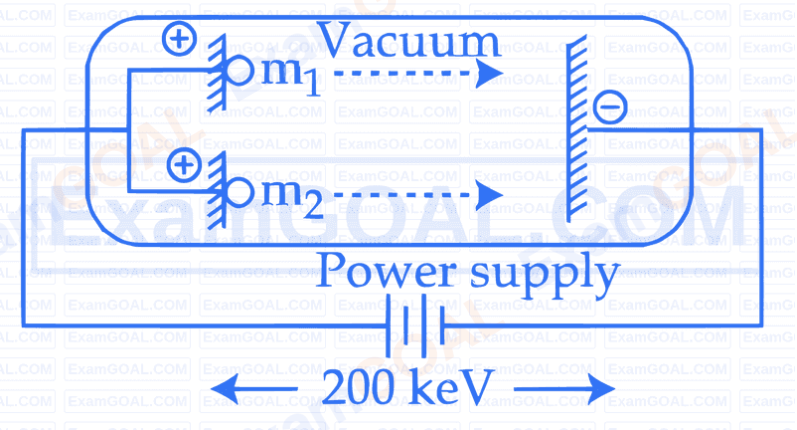
The surface of sodium metal is irradiated with radiation of wavelength x nm. The kinetic energy of ejected electrons is 2.8×10^{-20} J. The work function of sodium is 2.3 eV. The value of x is _____ $\times 10^2$ nm. (Nearest integer)

(Given: $h = 6.6 \times 10^{-34}$ J s ; 1 eV $= 1.6 \times 10^{-19}$ J ; $c = 3.0 \times 10^8$ m s^{-1})

Open solution

Q3. JEE Main 2026 (Online) 28th January Evening Shift

Two positively charged particles m_1 and m_2 have been accelerated across the same potential difference of 200 keV as shown below.



[Given mass of $m_1 = 1$ amu and $m_2 = 4$ amu]

The deBroglie wavelength of m_1 will be x times of m_2. The value of x is _____ (nearest integer)

Open solution

Q4. JEE Main 2026 (Online) 24th January Morning Shift

The hydrogen spectrum consists of several spectral lines in Lyman series ($L_1, L_2, \mathrm{L}_3 \ldots; \mathrm{L}_1$ has lowest energy among Lyman series). Similarly it consists of several spectral lines in Balmer series ($\mathrm{B}_1, \mathrm{B}_2, \mathrm{B}_3 \ldots; \mathrm{B}_1$ has lowest energy among Balmer lines). The energy of L_1 is x times the energy of B_1. The value of x is ____ $\times 10^{-1}$. (Nearest integer)

Open solution

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Q5. JEE Main 2025 (Online) 8th April Evening Shift

The energy of an electron in the first Bohr orbit of the H-atom is -13.6 eV.

The magnitude of energy value of an electron in the first excited state of Be^{3+} is _____ eV (nearest integer value).

Open solution

Q6. JEE Main 2024 (Online) 9th April Evening Shift

Based on Heisenberg's uncertainty principle, the uncertainty in the velocity of the electron to be found within an atomic nucleus of diameter $10^{-15}\ \mathrm{m}$ is _____ $\times 10^{9}\ \mathrm{ms}^{-1}$ (nearest integer)

[Given : mass of electron $= 9.1 \times 10^{-31}\ \mathrm{kg}$, Planck's constant $(h) = 6.626 \times 10^{-34}\mathrm{Js}$] (Value of $\pi = 3.14$)

Open solution

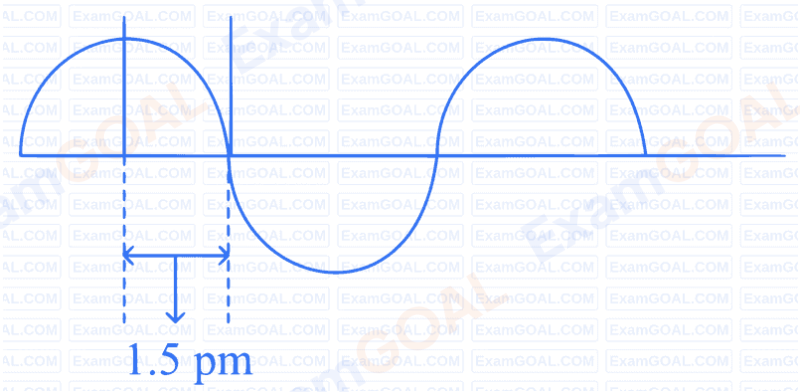
Q7. JEE Main 2024 (Online) 8th April Evening Shift

Wavenumber for a radiation having 5800 $\overset{o}{A}$ wavelength is $x \times 10\ \mathrm{cm}^{-1}$. The value of x is _____. (Integer answer)

Open solution

Q8. JEE Main 2024 (Online) 8th April Morning Shift

A hypothetical electromagnetic wave is shown below.



The frequency of the wave is $\mathrm{x} \times 10^{19}\ \mathrm{Hz}$.

$\mathrm{x} =$ _____ (nearest integer)

Open solution

Q9. JEE Main 2024 (Online) 6th April Evening Shift

For hydrogen atom, energy of an electron in first excited state is $-3.4\ \mathrm{eV}$. K.E. of the same electron of hydrogen atom is $x\ \mathrm{eV}$. Value of x is _____ $\times 10^{-1}\ \mathrm{eV}$. (Nearest integer)

Open solution

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Q10. JEE Main 2024 (Online) 6th April Morning Shift

Frequency of the de-Broglie wave of electron in Bohr's first orbit of hydrogen atom is _____ $\times 10^{13}$ Hz (nearest integer).

[Given : R_H (Rydberg constant) $= 2.18 \times 10^{-18}$ J, h (Planck's constant) $= 6.6 \times 10^{-34}$ J. s.]

Open solution

Q11. JEE Main 2024 (Online) 5th April Evening Shift

In an atom, total number of electrons having quantum numbers $n = 4, |m_l| = 1$ and $m_s = -\frac{1}{2}$ is _____.

Open solution

Q12. JEE Main 2024 (Online) 5th April Morning Shift

The value of Rydberg constant (R_H) is 2.18×10^{-18} J. The velocity of electron having mass 9.1×10^{-31} kg in Bohr's first orbit of hydrogen atom = _____ $\times 10^5$ ms^{-1} (nearest integer).

Open solution

Q13. JEE Main 2024 (Online) 4th April Evening Shift

The maximum number of orbitals which can be identified with $n = 4$ and $m_l = 0$ is _____.

Open solution

Q14. JEE Main 2024 (Online) 4th April Morning Shift

The de-Broglie's wavelength of an electron in the 4th orbit is _____ πa_0. (a_0 = Bohr's radius)

Open solution

Q15. JEE Main 2024 (Online) 31st January Morning Shift

The ionization energy of sodium in kJ mol^{-1}, if electromagnetic radiation of wavelength 242 nm is just sufficient to ionize sodium atom is _____.

Open solution

Q16. JEE Main 2024 (Online) 30th January Evening Shift

Number of spectral lines obtained in He^+ spectra, when an electron makes transition from fifth excited state to first excited state will be

Open solution

Q17. JEE Main 2024 (Online) 27th January Morning Shift

The number of electrons present in all the completely filled subshells having $n = 4$ and $s = +\frac{1}{2}$ is _____.

(Where n = principal quantum number and s = spin quantum number)

Open solution

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Q18. JEE Main 2023 (Online) 15th April Morning Shift

The total number of isoelectronic species from the given set is _____.

$O^{2-}, F^{-}, Al, Mg^{2+}, Na^{+}, O^{+}, Mg, Al^{3+}, F$

Open solution

Q19. JEE Main 2023 (Online) 13th April Evening Shift

The orbital angular momentum of an electron in $3\,s$ orbital is $\frac{xh}{2\pi}$. The value of x is _____ (nearest integer)

Open solution

Q20. JEE Main 2023 (Online) 12th April Morning Shift

Values of work function (W_0) for a few metals are given below

Metal	Li	Na	K	Mg	Cu	Ag
W_0/eV	2.42	2.3	2.25	3.7	4.8	4.3

The number of metals which will show photoelectric effect when light of wavelength $400\ nm$ falls on it is _____

Given: $h = 6.6 \times 10^{-34}\ J\,s$

$c = 3 \times 10^{8}\ ms^{-1}$

$e = 1.6 \times 10^{-19} C$

Open solution

Q21. JEE Main 2023 (Online) 11th April Evening Shift

The number of correct statements from the following is _____.

A. For $1s$ orbital, the probability density is maximum at the nucleus

B. For $2s$ orbital, the probability density first increases to maximum and then decreases sharply to zero.

C. Boundary surface diagrams of the orbitals encloses a region of 100% probability of finding the electron.

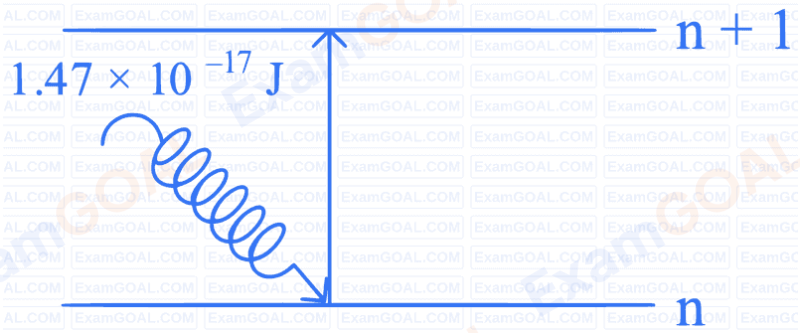
D. p and d-orbitals have 1 and 2 angular nodes respectively.

E. probability density of p-orbital is zero at the nucleus

Open solution

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Q22. JEE Main 2023 (Online) 10th April Evening Shift



Open solution

The electron in the nth orbit of Li^{2+} is excited to $(n+1)$ orbit using the radiation of energy 1.47×10^{-17} J (as shown in the diagram). The value of n is _____

Given: $R_H = 2.18 \times 10^{-18}$ J

Q23. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The number of incorrect statement/s about the black body from the following is _____

(A) Emit or absorb energy in the form of electromagnetic radiation.

(B) Frequency distribution of the emitted radiation depends on temperature.

(C) At a given temperature, intensity vs frequency curve passes through a maximum value.

(D) The maximum of the intensity vs frequency curve is at a higher frequency at higher temperature compared to that at lower temperature.

Q24. JEE Main 2023 (Online) 8th April Evening Shift

Open solution

The number of atomic orbitals from the following having 5 radial nodes is _____.

7s, 7p, 6s, 8p, 8d

Q25. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

The wavelength of an electron of kinetic energy 4.50×10^{-29} J is _____ $\times 10^{-5}$ m. (Nearest integer)

Given : mass of electron is 9×10^{-31} kg, h = 6.6×10^{-34} J s

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Q26. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

Electrons in a cathode ray tube have been emitted with a velocity of 1000 m s^{-1}. The number of following statements which is/are $\underline{\text{true}}$ about the emitted radiation is _____.

Given : $h = 6 \times 10^{-34}$ J s, $m_e = 9 \times 10^{-31}$ kg.

(A) The de-Broglie wavelength of the electron emitted is 666.67 nm.

(B) The characteristic of electrons emitted depend upon the material of the electrodes of the cathode ray tube.

(C) The cathode rays start from cathode and move towards anode.

(D) The nature of the emitted electrons depends on the nature of the gas present in cathode ray tube.

Q27. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

The energy of one mole of photons of radiation of frequency 2×10^{12} Hz in J mol^{-1} is _____. (Nearest integer)

[Given : $h = 6.626 \times 10^{-34}$ Js

$N_A = 6.022 \times 10^{23}$ mol^{-1}]

Q28. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Assume that the radius of the first Bohr orbit of hydrogen atom is 0.6 $\mathring{A}$. The radius of the third Bohr orbit of He^+ is _____ picometer. (Nearest Integer)

Q29. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

The number of given orbitals which have electron density along the axis is _____

$p_x, p_y, p_z, d_{xy}, d_{yz}, d_{xz}, d_{z^2}, d_{x^2-y^2}$

Q30. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

If wavelength of the first line of the Paschen series of hydrogen atom is 720 nm, then the wavelength of the second line of this series is _____ nm. (Nearest integer)

Q31. JEE Main 2022 (Online) 29th July Morning Shift

Open solution

The minimum uncertainty in the speed of an electron in an one dimensional region of length $2a_o$ (Where a_o = Bohr radius 52.9 pm) is _____ km s^{-1}.

(Given : Mass of electron = 9.1 $\times$ 10^{-31} kg, Planck's constant h = 6.63 $\times$ 10^{-34} Js)

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Q32. JEE Main 2022 (Online) 28th July Evening Shift

If the wavelength for an electron emitted from H-atom is 3.3×10^{-10} m, then energy absorbed by the electron in its ground state compared to minimum energy required for its escape from the atom, is _____ times. (Nearest integer)

[Given : $h = 6.626 \times 10^{-34}$ J s]

Mass of electron $= 9.1 \times 10^{-31}$ kg

Open solution

Q33. JEE Main 2022 (Online) 26th July Evening Shift

Consider an imaginary ion $^{48}_{22}X^{3-}$. The nucleus contains 'a'% more neutrons than the number of electrons in the ion. The value of 'a' is _____. [nearest integer]

Open solution

Q34. JEE Main 2022 (Online) 26th July Morning Shift

The wavelength of an electron and a neutron will become equal when the velocity of the electron is x times the velocity of neutron. The value of x is _____. (Nearest Integer)

(Mass of electron is 9.1×10^{-31} kg and mass of neutron is 1.6×10^{-27} kg)

Open solution

Q35. JEE Main 2022 (Online) 25th July Evening Shift

When the excited electron of a H atom from n = 5 drops to the ground state, the maximum number of emission lines observed are _____.

Open solution

Q36. JEE Main 2022 (Online) 28th June Morning Shift

If the work function of a metal is 6.63 $\times$ 10^{-19}J, the maximum wavelength of the photon required to remove a photoelectron from the metal is _____ nm. (Nearest integer)

[Given : h = 6.63 $\times$ 10^{-34} J s, and c = 3 $\times$ 10^{8} m s^{-1}]

Open solution

Q37. JEE Main 2022 (Online) 27th June Evening Shift

Consider the following set of quantum numbers.

	n	l	m_l
A.	3	3	-3
B.	3	2	-2
C.	2	1	+1
D.	2	2	+2

The number of correct sets of quantum numbers is _____.

Open solution

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Q38. JEE Main 2022 (Online) 27th June Morning Shift

If the uncertainty in velocity and position of a minute particle in space are, 2.4×10^{-26} (m s^{-1}) and 10^{-7} (m) respectively. The mass of the particle in g is _____. (Nearest integer)

(Given : h = 6.626×10^{-34} Js)

Open solution

Q39. JEE Main 2022 (Online) 25th June Morning Shift

The longest wavelength of light that can be used for the ionisation of lithium atom (Li) in its ground state is $x \times 10^{-8}$ m. The value of x is _____. (Nearest Integer).

(Given : Energy of the electron in the first shell of the hydrogen atom is -2.2×10^{-18} J ; h = 6.63×10^{-34} Js and c = 3×10^{8} ms^{-1})

Open solution

Q40. JEE Main 2021 (Online) 1st September Evening Shift

A 50 watt bulb emits monochromatic red light of wavelength of 795 nm. The number of photons emitted per second by the bulb is $x \times 10^{20}$. The value of x is _____. [Given : h = 6.63×10^{-34} Js and c = 3.0×10^{8} ms^{-1}]

Open solution

Q41. JEE Main 2021 (Online) 31st August Morning Shift

Ge(Z = 32) in its ground state electronic configuration has x completely filled orbitals with $m_l = 0$. The value of x is _____.

Open solution

Q42. JEE Main 2021 (Online) 27th August Evening Shift

The number of photons emitted by a monochromatic (single frequency) infrared range finder of power 1 mW and wavelength of 1000 nm, in 0.1 second is $x \times 10^{13}$. The value of x is _____. (Nearest integer)

(h = 6.63×10^{-34} Js, c = 3.00×10^{8} ms^{-1})

Open solution

Q43. JEE Main 2021 (Online) 27th August Morning Shift

The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is equal to $\frac{h^2}{xma_0^2}$. The value of 10x is _____. (a_0 is radius of Bohr's orbit) (Nearest integer) [Given : π = 3.14]

Open solution

Q44. JEE Main 2021 (Online) 26th August Evening Shift

A metal surface is exposed to 500 nm radiation. The threshold frequency of the metal for photoelectric current is 4.3×10^{14} Hz. The velocity of ejected electron is _____ $\times 10^{5}$ ms^{-1} (Nearest integer)

[Use : h = 6.63×10^{-34} Js, m_e = 9.0×10^{-31} kg]

Open solution

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Q45. JEE Main 2021 (Online) 25th July Evening Shift

An accelerated electron has a speed of 5×10^6 ms^{-1} with an uncertainty of 0.02%. The uncertainty in finding its location while in motion is $x \times 10^{-9}$ m. The value of x is _____. (Nearest integer)

[Use mass of electron = 9.1×10^{-31} kg, h =6.63×10^{-34} Js, π = 3.14]

Open solution

Q46. JEE Main 2021 (Online) 25th July Morning Shift

A source of monochromatic radiation of wavelength 400 nm provides 1000 J of energy in 10 seconds. When this radiation falls on the surface of sodium, $x \times 10^{20}$ electrons are ejected per second. Assume that wavelength 400 nm is sufficient for ejection of electron from the surface of sodium metal. The value of x is _____. (Nearest integer)

(h = 6.626×10^{-34} Js)

Open solution

Q47. JEE Main 2021 (Online) 20th July Evening Shift

The wavelength of electrons accelerated from rest through a potential difference of 40 kV is $x \times 10^{-12}$ m. The value of x is _____. (Nearest integer)

Given : Mass of electron = 9.1×10^{-31} kg

Charge on an electron = 1.6×10^{-19} C

Planck's constant = 6.63×10^{-34} Js

Open solution

Q48. JEE Main 2021 (Online) 20th July Morning Shift

The Azimuthal quantum number for the valence electrons of Ga^+ ion is _____.

(Atomic number of Ga = 31)

Open solution

Q49. JEE Main 2021 (Online) 17th March Morning Shift

A certain orbital has n = 4 and $m_L = -3$. The number of radial nodes in this orbital is _____. (Round off to the Nearest Integer).

Open solution

Q50. JEE Main 2021 (Online) 16th March Evening Shift

The number of orbitals with n = 5, m_l = +2 is _____. (Round off to the Nearest Integer).

Open solution

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Q51. JEE Main 2021 (Online) 16th March Morning Shift

When light of wavelength 248 nm falls on a metal of threshold energy 3.0 eV, the de-Broglie wavelength of emitted electrons is _____ $\overset{o}{A}$. (Round off to the Nearest Integer).

[Use : $\sqrt{3}$ = 1.73, h = 6.63×10^{-34} Js

$m_e = 9.1 \times 10^{-31}$ kg; c = 3.0×10^8 ms^{-1}; 1eV = 1.6×10^{-19} J]

Open solution

Q52. JEE Main 2021 (Online) 26th February Evening Shift

A ball weighing 10 g is moving with a velocity of 90 ms^{-1}. If the uncertainty in its velocity is 5%, then the uncertainty in its position is _____ $\times 10^{-33}$ m. (Rounded off to the nearest integer)

[Given : h = 6.63×10^{-34} Js]

Open solution

Q53. JEE Main 2021 (Online) 25th February Evening Shift

Electromagnetic radiation of wavelength 663 nm is just sufficient to ionise the atom of metal A. The ionization energy of metal A in kJ mol^{-1} is _____. (Rounded off to the nearest integer)

[h = 6.63×10^{-34} Js, c = 3.00×10^8 ms^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1}]

Open solution

Q54. JEE Main 2021 (Online) 25th February Evening Shift

Among the following, number of metal/s which can be used as electrodes in the photoelectric cell is _____. (Integer answer)

(A) Li

(B) Na

(C) Rb

(D) Cs

Open solution

Q55. JEE Main 2021 (Online) 24th February Morning Shift

A proton and a Li^{3+} nucleus are accelerated by the same potential. If λ_{Li} and λ_p denote the de Broglie wavelengths of Li^{3+} and proton respectively, then the value of $\frac{\lambda_{Li}}{\lambda_p}$ is x $\times 10^{-1}$.
The value of x is _____. (Rounded off to the nearest integer)
[Mass of Li^{3+} = 8.3 mass of proton]

Open solution

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Q56. JEE Main 2020 (Online) 2nd September Evening Slot

The work function of sodium metal is 4.41 $\times$ 10^{-19} J. If photons of wavelength 300 nm are incident on the metal, the kinetic energy of the ejected electrons will be (h = 6.63 $\times$ 10^{-34} J s; c = 3 $\times$ 10^{8} m/s) _____ $\times$ 10^{-21} J.

Open solution

MCQ (Single Correct Answer) 173 questions

Q57. JEE Main 2026 (Online) 6th April Evening Shift

The Bohr radius of a hydrogen like species is 70.53 pm . The species and the stationary state (n) are respectively

(Given : Hydrogen atom Bohr radius is 52.9 pm)

Open solution

(A) $\mathrm{Li}^{2+}, 3$ (B) $\mathrm{He}^{+}, 3$ (C) $\mathrm{He}^{+}, 2$ (D) $\mathrm{Li}^{2+}, 2$

Q58. JEE Main 2026 (Online) 6th April Morning Shift

If shortest wavelength of hydrogen atom in Lyman series is x, then longest wavelength in Balmer series of $\mathrm{He}^{+}$is:

Open solution

(A) $\frac{9x}{5}$ (B) $\frac{36x}{5}$ (C) $\frac{x}{4}$ (D) $\frac{5x}{9}$

Q59. JEE Main 2026 (Online) 6th April Morning Shift

Match the LIST-I with LIST-II

Open solution

List-I Orbital		List-II Radial nodes and nodal plane	
A.	2s	I.	1 Radial node + two nodal planes
B.	3s	II.	1 Radial node + one nodal plane
C.	3p	III.	2 Radial nodes + No nodal plane
D.	4d	IV.	1 Radial node + No nodal plane

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II (B) A-IV, B-II, C-III, D-I

(C) A-III, B-I, C-IV, D-II (D) A-IV, B-III, C-II, D-I

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Q60. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

Which of the following statement(s) is/are true ?

A. If two orbitals have the same value of ($n + l$), the orbital with lower value of n will have lower energy.

B. Energies of the orbitals in the same subshell increase with increase in atomic number.

C. The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.

D. Among 5f, 6s, 4d, 5p and 5d orbitals, none of the orbitals have 2 radial nodes.

Choose the correct answer from the options given below :

(A) A, B and C only **(B)** A and C only **(C)** C and D only **(D)** A only

Q61. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Arrange the following atomic orbitals of multi electron atoms in order of increasing energy.

A. $n = 3, l = 2, m = +1$

B. $n = 4, l = 0, m = 0$

C. $n = 6, l = 1, m = 0$

D. $n = 5, l = 1, m = +1$

E. $n = 2, l = 1, m = +1$

Choose the correct answer from the options given below:

(A) $C < D < B < A < E$ **(B)** $B < A < E < C < D$

(C) $E < C < D < B < A$ **(D)** $E < B < A < D < C$

Q62. JEE Main 2026 (Online) 5th April Morning Shift

Open solution

Identify the correct statements from the following :

A. Heisenberg uncertainty principle is applicable to electrons.

B. The size of $2p_x$ orbital is less than the size of $3p_x$ orbital.

C. The energy of 2 s orbital of H atom is equal to the energy of 2 s orbital of Li .

D. The electronic configuration of Cr is $[Ar]3\,d^5 4\,s^1$

Choose the correct answer from the options given below:

(A) A, B and C Only **(B)** A, B and D Only **(C)** B, C and D Only **(D)** A, C and D Only

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Q63. JEE Main 2026 (Online) 4th April Evening Shift

The species having identical radii according to the Bohr's theory are :

A. H (first orbit)

B. He^{+}(first orbit)

C. He^{+}(Second orbit)

D. Li^{2+} (first orbit)

E. Be^{3+} (Second orbit)

Choose the correct answer from the options given below :

(A) A and C Only (B) A and E Only (C) B and E Only (D) C and D Only

Open solution

Q64. JEE Main 2026 (Online) 4th April Morning Shift

What is the ratio of wave number of first line (lowest energy line) of Balmer series of H atomic spectrum to first line of its Brackett series?

(A) 5 : 1 (B) 5 : 0.81 (C) 5 : 1.75 (D) 5 : 27

Open solution

Q65. JEE Main 2026 (Online) 4th April Morning Shift

Which of the following is correct set of 4 quantum numbers of 19th electron in Chromium (Atomic number $= 24$) in accordance with Aufbau principle?

(A) $n = 3, l = 2, m = +2, s = +\frac{1}{2}$ (B) $n = 3, l = 2, m = -2, s = +\frac{1}{2}$

(C) $n = 4, l = 1, m = 0, s = +\frac{1}{2}$ (D) $n = 4, l = 0, m = 0, s = +\frac{1}{2}$

Open solution

Q66. JEE Main 2026 (Online) 2nd April Morning Shift

What is the energy (in J atom^{-1}) required for the following process?

$Li^{2+}(g) \rightarrow Li^{3+}(g) + e^{-}$

(Take the ionization energy for the H atom in the ground state as 2.18×10^{-18} J atom^{-1})

(A) 8.72×10^{-18} (B) 1.962×10^{-18} (C) 1.962×10^{-17} (D) 6.54×10^{-17}

Open solution

Q67. JEE Main 2026 (Online) 28th January Evening Shift

The wavelength of photon 'A' is 400 nm. The frequency of photon 'B' is 10^{16} s^{-1}. The wave number of photon 'C' is 10^{4} cm^{-1}. The correct order of energy of these photons is :

(A) A > C > B (B) C > B > A (C) B > A > C (D) A > B > C

Open solution

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Q68. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

The wave numbers of three spectral lines of H atom are considered. Identify the set of spectral lines belonging to Balmer series.

(R = Rydberg constant)

(A) $\frac{5R}{36}, \frac{3R}{16}, \frac{21R}{100}$

(B) $\frac{7R}{144}, \frac{3R}{16}, \frac{16R}{255}$

(C) $\frac{3R}{4}, \frac{3R}{16}, \frac{7R}{144}$

(D) $\frac{5R}{36}, \frac{8R}{9}, \frac{15R}{16}$

Q69. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

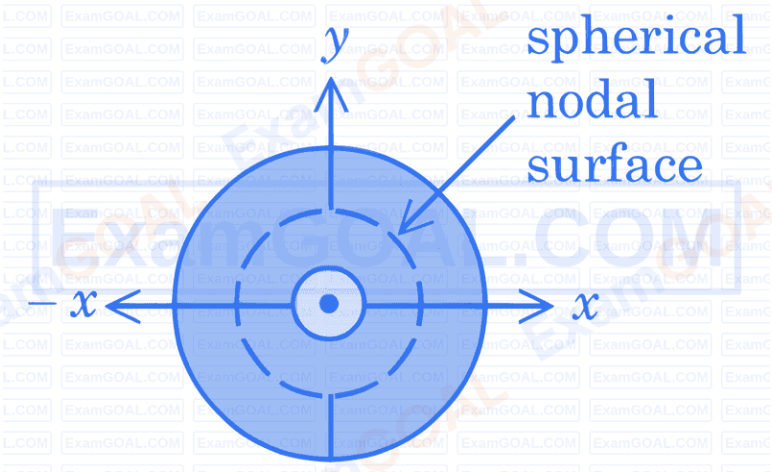


Figure 1. electron probability density for 2 s orbital

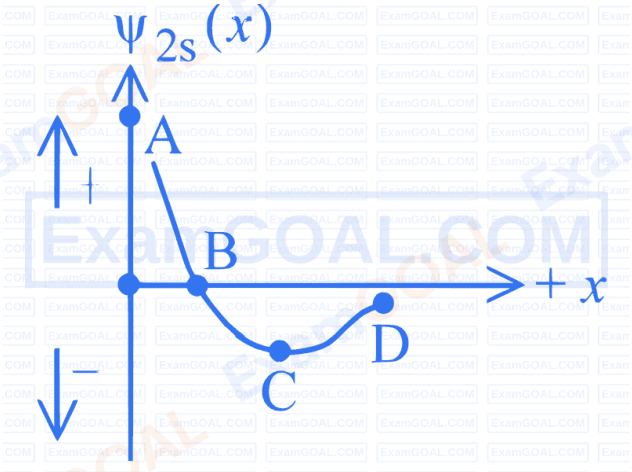


Figure 2. wave function for 2s orbital

Which of the following point in Figure 2 most accurately represents the nodal surface as shown in Figure 1?

(A) D (B) B (C) C (D) A

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Q70. JEE Main 2026 (Online) 24th January Evening Shift

The wavelength of spectral line obtained in the spectrum of Li^{2+} ion, when the transition takes place between two levels whose sum is 4 and difference is 2 , is

(A) 1.14×10^{-6} cm (B) 2.28×10^{-6} cm (C) 1.14×10^{-7} cm (D) 2.28×10^{-7} cm

Open solution

Q71. JEE Main 2026 (Online) 23rd January Evening Shift

The work functions of two metals $(M_A$ and $M_B)$ are in the $1:2$ ratio. When these metals are exposed to photons of energy 6 eV , the kinetic energy of liberated electrons of $M_A : M_B$ is in the ratio of $2.642:1$. The work functions (in eV) of M_A and M_B are respectively.

(A) $3.1, 6.2$ (B) $1.5, 3.0$ (C) $2.3, 4.6$ (D) $1.4, 2.8$

Open solution

Q72. JEE Main 2026 (Online) 23rd January Evening Shift

Identify the **INCORRECT** statements from the following :

A. Notation $^{24}_{12}Mg$ represents 24 protons and 12 neutrons.

B. Wavelength of a radiation of frequency $4.5 \times 10^{15}\ s^{-1}$ is 6.7×10^{-8} m.

C. One radiation has wavelength $= \lambda_1(900\ nm)$ and energy $= E_1$. Other radiation has wavelength $= \lambda_2(300\ nm)$ and energy $= E_2$. $E_1 : E_2 = 3:1$.

D. Number of photons of light of wavelength 2000 pm that provides 1 J of energy is 1.006×10^{16}.

Choose the correct answer from the options given below :

(A) A and D Only (B) A and C Only (C) A and B Only (D) B and C Only

Open solution

Q73. JEE Main 2026 (Online) 23rd January Morning Shift

Given,

(A) $n = 5,\ m_l = -1$

(B) $n = 3, l = 2,\ m_l = -1,\ m_s = +\frac{1}{2}$

The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) respectively are :

(A) 4 and 1 (B) 2 and 4 (C) 26 and 1 (D) 8 and 1

Open solution

Q74. JEE Main 2026 (Online) 23rd January Morning Shift

Which of the following statements regarding the energy of the stationary state is true in the following one - electron systems ?

(A) $+2.18 \times 10^{-18}$ J for second orbit of He^+ion

(B) $+8.72 \times 10^{-18}$ J for first orbit of He^+ion

(C) -1.09×10^{-18} J for second orbit of H atom.

(D) -2.18×10^{-18} J for third orbit of Li^{2+} ion

Open solution

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Q75. JEE Main 2026 (Online) 22nd January Evening Shift

The energy of first (lowest) Balmer line of H atom is x J. The energy (in J) of second Balmer line of H atom is :

(A) $\frac{x}{1.35}$ (B) $1.35x$ (C) x^2 (D) $2x$

Open solution

Q76. JEE Main 2026 (Online) 22nd January Morning Shift

The energy required by electrons, present in the first Bohr orbit of hydrogen atom to be excited to second Bohr orbit is ____ Jmol^{-1}.

Given: $R_H = 2.18 \times 10^{-11}\text{ergs}$.

(A) 9.835×10^{12} (B) 1.635×10^{-11} (C) 1.635×10^{-18} (D) 9.835×10^{5}

Open solution

Q77. JEE Main 2026 (Online) 21st January Evening Shift

Consider the following spectral lines for atomic hydrogen :

A. First line of Paschen series

B. Second line of Balmer series

C. Third line of Paschen series

D. Fourth line of Brackett series

The correct arrangement of the above lines in ascending order of energy is :

(A) D < C < A < B (B) A < B < C < D (C) D < A < C < B (D) C < D < B < A

Open solution

Q78. JEE Main 2026 (Online) 21st January Morning Shift

Given below are two statements :

Statement I : When an electric discharge is passed through gaseous hydrogen, the hydrogen molecules dissociate and the energetically excited hydrogen atoms produce electromagnetic radiation of discrete frequencies.

Statement II : The frequency of second line of Balmer series obtained from He^+ is equal to that of first line of Lyman series obtained from hydrogen atom.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Open solution

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Q79. JEE Main 2025 (Online) 8th April Evening Shift

Open solution

Correct statements for an element with atomic number 9 are:

A. There can be 5 electrons for which $m_s = +\frac{1}{2}$ and 4 electrons for which $m_s = -\frac{1}{2}$.

B. There is only one electron in p_z orbital.

C. The last electron goes to orbital with $n = 2$ and $l = 1$.

D. The sum of angular nodes of all the atomic orbitals is 1.

Choose the correct answer from the options given below:

(A) A and B Only **(B)** A, C and D Only **(C)** A and C Only **(D)** C and D Only

Q80. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

The extra stability of half-filled subshell is due to :

(A) Symmetrical distribution of electrons

(B) Smaller coulombic repulsion energy

(C) The presence of electrons with the same spin in non-degenerate orbitals

(D) Larger exchange energy

(E) Relatively smaller shielding of electrons by one another

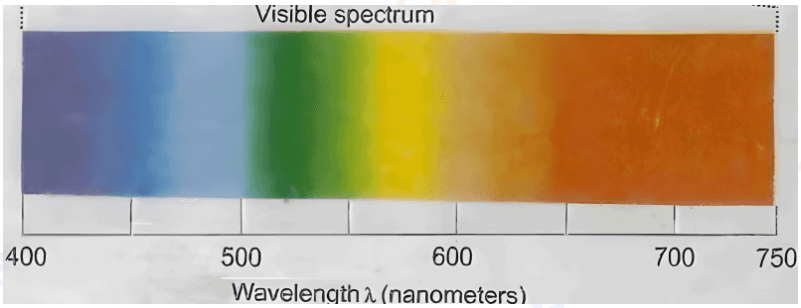
Indentify the correct statements :

(A) (B), (D) and (E) only **(B)** $(A), (B), (D)$ and (E) only

(C) (B), (C) and (D) only **(D)** $(A), (B)$ and (D) only

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Q81. JEE Main 2025 (Online) 7th April Morning Shift



Open solution

Which of the following statements are correct, if the threshold frequency of caesium is 5.16×10^{14} Hz ?

A. When Cs is placed inside a vacuum chamber with an ammeter connected to it and yellow light is focused on Cs , the ammeter shows the presence of current.

B. When the brightness of the yellow light is dimmed, the value of the current in the ammeter is reduced.

C. When a red light is used instead of the yellow light, the current produced is higher with respect to the yellow light.

D. When a blue light is used, the ammeter shows the formation of current.

E. When a white light is used, the ammeter shows formation of current.

Choose the correct answer from the options given below:

(A) B, C and D Only
(B) A, C, D and E Only
(C) A, B, D and E Only
(D) A, D and E Only

Q82. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Consider the ground state of chromium atom $(Z = 24)$. How many electrons are with Azimuthal quantum number $l = 1$ and $l = 2$ respectively ?

(A) 16 and 5
(B) 12 and 5
(C) 12 and 4
(D) 16 and 4

Q83. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Which one of the following about an electron occupying the 1 s orbital in a hydrogen atom is incorrect?

(Bohr's radius is represented by a_0)

(A) The probability density of finding the electron is maximum at the nucleus
(B) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus
(C) The electron can be found at a distance $2a_0$ from the nucleus
(D) The 1 s orbital is spherically symmetrical

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Q84. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

For electrons in ' 2 s ' and ' 2 p ' orbitals, the orbital angular momentum values, respectively are:

(A) 0 and $\sqrt{6}\frac{h}{2\pi}$

(B) 0 and $\sqrt{2}\frac{h}{2\pi}$

(C) $\sqrt{2}\frac{h}{2\pi}$ and 0

(D) $\frac{h}{2\pi}$ and $\sqrt{2}\frac{h}{2\pi}$

Q85. JEE Main 2025 (Online) 3rd April Morning Shift

Open solution

Which of the following postulate of Bohr's model of hydrogen atom is not in agreement with quantum mechanical model of an atom?

(A) The electron in a H atom's stationary state moves in a circle around the nucleus.

(B) An atom can take only certain distinct energies $\mathrm{E}_1, \mathrm{E}_2, \mathrm{E}_3$, etc. These allowed states of constant energy are called the stationary states of atom.

(C) An atom in a stationary state does not emit electromagnetic radiation as long as it stays in the same state.

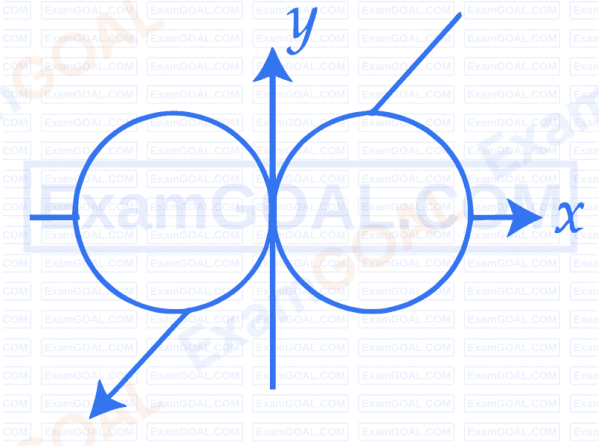
(D) When an electron makes a transition from a higher energy stationary state to a lower energy stationary state, then it emits a photon of light.

Q86. JEE Main 2025 (Online) 2nd April Evening Shift

Open solution

Which of the following statements are true?

(A) The subsidiary quantum number l describes the shape of the orbital occupied by the electron.

(B) 

is the boundary surface diagram of the $2\mathrm{p}_x$ orbital.

(C) The + and - signs in the wave function of the $2p_x$ orbital refer to charge.

(D) The wave function of $2\mathrm{p}_x$ orbital is zero everywhere in the xy plane.

Choose the correct answer from the options given below :

(A) (A) and (B) only

(B) (B) and (D) only

(C) (C) and (D) only

(D) (A), (B) and (C) only

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Q87. JEE Main 2025 (Online) 2nd April Morning Shift

Open solution

According to Bohr's model of hydrogen atom, which of the following statement is incorrect?

(A) Radius of 4th orbit is four times larger than that of 2nd orbit

(B) Radius of 8th orbit is four times larger than that of 4th orbit

(C) Radius of 6th orbit is three times larger than that of 4th orbit

(D) Radius of 3rd orbit is nine times larger than that of 1st orbit

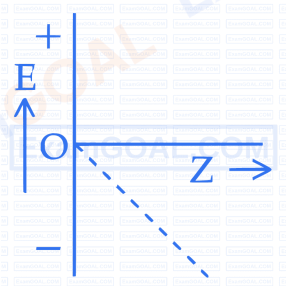
Q88. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

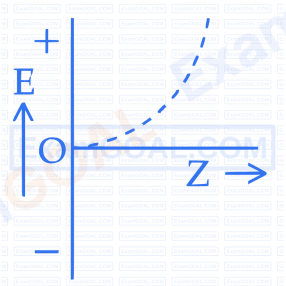
For hydrogen like species, which of the following graphs provides the most appropriate representation of E vs Z plot for a constant n?

[E : Energy of the stationary state, Z : atomic number, n = principal quantum number]

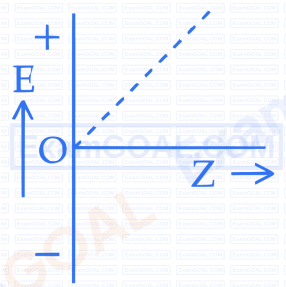
(A)



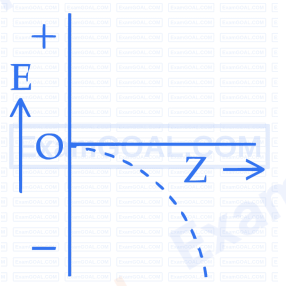
(B)



(C)



(D)



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Q89. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

Given below are two statements :

Statement (I): It is impossible to specify simultaneously with arbitrary precision, both the linear momentum and the position of a particle.

Statement (II) : If the uncertainty in the measurement of position and uncertainty in measurement of momentum are equal for an electron, then the uncertainty in the measurement of velocity is $\geqslant \sqrt{\frac{h}{\pi}} \times \frac{1}{2m}$.

In the light of the above statements, choose the correct answer from the options given below :

(A) **Statement I** is false but **Statement II** is true

(B) Both **Statement I** and **Statement II** are true

(C) Both **Statement I** and **Statement II** are false

(D) **Statement I** is true but **Statement II** is false

Q90. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

If a_0 is denoted as the Bohr radius of hydrogen atom, then what is the de-Broglie wavelength (λ) of the electron present in the second orbit of hydrogen atom? [n : any integer]

(A) $\frac{8\pi a_0}{n}$ (B) $\frac{2a_0}{n\pi}$ (C) $\frac{na_0}{4\pi}$ (D) $\frac{4\pi a_0}{n}$

Q91. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

Which of the following is/are not correct with respect to energy of atomic orbitals of hydrogen atom?

(A) 1s < 2p < 3d < 4s

(B) 1s < 2s = 2p < 3s = 3p

(C) 1s < 2s < 2p < 3s < 3p

(D) 1s < 2s < 4s < 3d

Choose the **correct** answer from the options given below :

(A) (A) and (B) only (B) (A) and (C) only (C) (B) and (D) only (D) (C) and (D) only

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Q92. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

In a multielectron atom, which of the following orbitals described by three quantum numbers will have same energy in absence of electric and magnetic fields?

A. $\mathrm{n}=1, \mathrm{l}=0,\ \mathrm{m}_l=0$

B. $\mathrm{n}=2, \mathrm{l}=0,\ \mathrm{m}_l=0$

C. $\mathrm{n}=2, \mathrm{l}=1,\ \mathrm{m}_l=1$

D. $\mathrm{n}=3, \mathrm{l}=2,\ \mathrm{m}_l=1$

E. $\mathrm{n}=3, \mathrm{l}=2,\ \mathrm{m}_l=0$

Choose the correct answer from the options given below:

(A) B and C Only **(B)** C and D Only **(C)** A and B Only **(D)** D and E Only

Q93. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

For hydrogen atom, the orbital/s with lowest energy is/are :

(A) $4\mathrm{s}$

(B) $3\mathrm{p}_x$

(C) $3\ \mathrm{d}_{x^2-y^2}$

(D) $3\ \mathrm{d}_{z^2}$

(E) $4\mathrm{p}_z$

Choose the correct answer from the options given below :

(A) (A) only **(B)** (B) only

(C) (B), (C) and (D) only **(D)** (A) and (E) only

Q94. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Given below are two statements :

Statement (I) : For a given shell, the total number of allowed orbitals is given by n^2.

Statement (II) : For any subshell, the spatial orientation of the orbitals is given by $-l$ to $+l$ values including zero.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

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Q95. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

Given below are two statements about X-ray spectra of elements :

Statement (I) : A plot of $\sqrt{\nu}$ ($\nu =$ frequency of X-rays emitted) vs atomic mass is a straight line.

Statement (II) : A plot of ν ($\nu =$ frequency of X-rays emitted) vs atomic number is a straight line. In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

Q96. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Heat treatment of muscular pain involves radiation of wavelength of about 900 nm . Which spectral line of H atom is suitable for this?

Given : Rydberg constant $\mathrm{R_H} = 10^5\ \mathrm{cm^{-1}}$, $\mathrm{h} = 6.6 \times 10^{-34}\ \mathrm{J\ s}, \mathrm{c} = 3 \times 10^8\ \mathrm{m/s}\Big)$

(A) Lyman series, $\infty \rightarrow 1$

(B) Balmer series, $\infty \rightarrow 2$

(C) Paschen series, $5 \rightarrow 3$

(D) Paschen series, $\infty \rightarrow 3$

Q97. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Given below are two statements :

Statement (I) : A spectral line will be observed for a $2p_x \rightarrow 2p_y$ transition.

Statement (II) : $2\mathrm{p}_x$ and $2\mathrm{p}_y$ are degenerate orbitals.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are true

Q98. JEE Main 2025 (Online) 22nd January Morning Shift

Open solution

Radius of the first excited state of Helium ion is given as : $\mathrm{a}_0 \rightarrow$ radius of first stationary state of hydrogen atom.

(A) $\mathrm{r} = \frac{\mathrm{a}_0}{4}$

(B) $\mathrm{r} = 2\mathrm{a}_0$

(C) $\mathrm{r} = 4\mathrm{a}_0$

(D) $\mathrm{r} = \frac{\mathrm{a}_0}{2}$

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Q99. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

Compare the energies of following sets of quantum numbers for multielectron system.

(A) $\mathrm{n}=4, \mathrm{l}=1$

(B) $\mathrm{n}=4, \mathrm{l}=2$

(C) $\mathrm{n}=3, \mathrm{l}=1$

(D) $\mathrm{n}=3, \mathrm{l}=2$

(E) $\mathrm{n}=4, \mathrm{l}=0$

Choose the correct answer from the options given below :

(A) $(\mathrm{E})>(\mathrm{C})>(\mathrm{A})>(\mathrm{D})>(\mathrm{B})$

(B) $(\mathrm{B})>(\mathrm{A})>(\mathrm{C})>(\mathrm{E})>(\mathrm{D})$

(C) $(\mathrm{C})<(\mathrm{E})<(\mathrm{D})<(\mathrm{A})<(\mathrm{B})$

(D) $(\mathrm{E})<(\mathrm{C})<(\mathrm{D})<(\mathrm{A})<(\mathrm{B})$

Q100. JEE Main 2024 (Online) 4th April Evening Shift

Open solution

Choose the Incorrect Statement about Dalton's Atomic Theory

(A) Compounds are formed when atoms of different elements combine in any ratio.

(B) Matter consists of indivisible atoms.

(C) chemical reactions involve reorganization of atoms

(D) All the atoms of a given element have identical properties including identical mass.

Q101. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

The number of radial node/s for $3p$ orbital is :

(A) 3 (B) 2 (C) 1 (D) 4

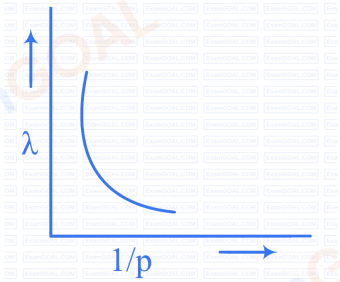
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Q102. JEE Main 2024 (Online) 1st February Morning Shift

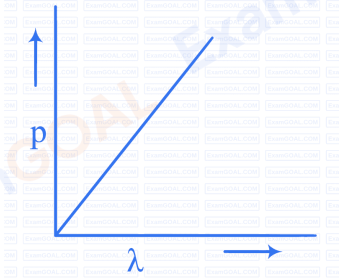
Open solution

According to the wave-particle duality of matter by de-Broglie, which of the following graph plot presents most appropriate relationship between wavelength of electron (λ) and momentum of electron (p)?

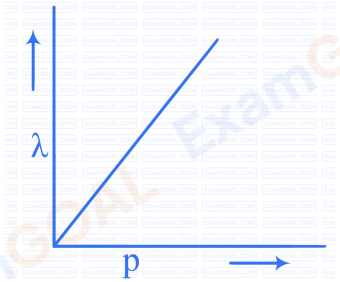
(A)



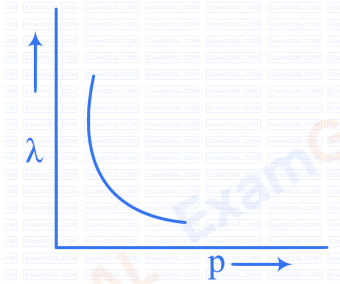
(B)



(C)



(D)



Q103. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

The four quantum numbers for the electron in the outermost orbital of potassium (atomic no. 19) are

(A) $\mathrm{n}=3, l=0, \mathrm{m}=1, \mathrm{s}=+\frac{1}{2}$

(B) $\mathrm{n}=4, l=0, \mathrm{m}=0, s=+\frac{1}{2}$

(C) $\mathrm{n}=2, l=0, \mathrm{m}=0, s=+\frac{1}{2}$

(D) $\mathrm{n}=4, l=2, \mathrm{m}=-1, s=+\frac{1}{2}$

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Q104. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Given below are two statements :

Statement (I) : The orbitals having same energy are called as degenerate orbitals.

Statement (II) : In hydrogen atom, 3p and 3d orbitals are not degenerate orbitals.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q105. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

Match List I with List II

	List - I (Spectral Series for Hydrogen)		**List - II (Spectral Region/Higher Energy State)**
(A)	Lyman	(I)	Infrared region
(B)	Balmer	(II)	UV region
(C)	Paschen	(III)	Infrared region
(D)	Pfund	(IV)	Visible region

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-III, D-I

(B) A-I, B-III, C-II, D-IV

(C) A-II, B-III, C-I, D-IV

(D) A-I, B-II, C-III, D-IV

Q106. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

The correct set of four quantum numbers for the valence electron of rubidium atom $(Z = 37)$ is :

(A) $5, 1, 1, +\frac{1}{2}$ **(B)** $5, 0, 0, +\frac{1}{2}$ **(C)** $5, 0, 1, +\frac{1}{2}$ **(D)** $5, 1, 0, +\frac{1}{2}$

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Q107. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

Given below are two statements:

Statement I : According to Bohr's model of hydrogen atom, the angular momentum of an electron in a given stationary state is quantised.

Statement II : The concept of electron in Bohr's orbit, violates the Heisenberg uncertainty principle.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are incorrect

(B) Statement I is correct but Statement II is incorrect

(C) Both Statement I and Statement II are correct

(D) Statement I is incorrect but Statement II is correct

Q108. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

The energy of an electron in the first Bohr orbit of hydrogen atom is -2.18×10^{-18} J. Its energy in the third Bohr orbit is _____.

(A) One third of this value

(B) Three times of this value

(C) $\frac{1}{9}$ th of this value

(D) $\frac{1}{27}$ of this value

Q109. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : In the photoelectric effect, the electrons are ejected from the metal surface as soon as the beam of light of frequency greater than threshold frequency strikes the surface.

Reason R : When the photon of any energy strikes an electron in the atom, transfer of energy from the photon to the electron takes place.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are correct and R is the correct explanation of A

(B) A is not correct but R is correct

(C) A is correct but R is not correct

(D) Both A and R are correct but R is NOT the correct explanation of A

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Q110. JEE Main 2023 (Online) 8th April Evening Shift

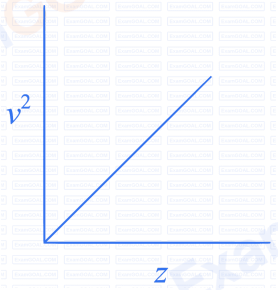
Open solution

Henry Moseley studied characteristic X-ray spectra of elements. The graph which represents his observation correctly is

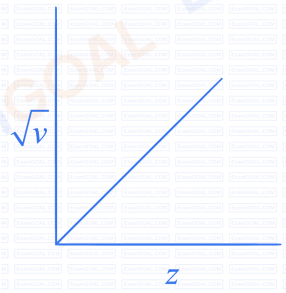
Given $v =$ frequency of X-ray emitted

Z = atomic number

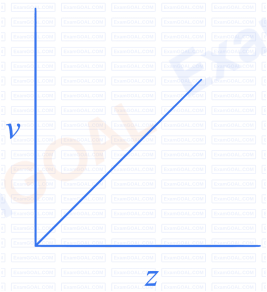
(A)



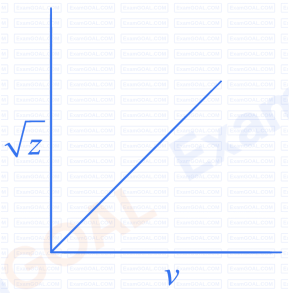
(B)



(C)



(D)



Q111. JEE Main 2023 (Online) 6th April Evening Shift

Open solution

If the radius of the first orbit of hydrogen atom is α_0, then de Broglie's wavelength of electron in 3rd orbit is :

(A) $\frac{\pi\alpha_0}{6}$ (B) $3\pi\alpha_0$ (C) $6\pi\alpha_0$ (D) $\frac{\pi\alpha_0}{3}$

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Q112. JEE Main 2023 (Online) 1st February Evening Shift

Which one of the following sets of ions represents a collection of isoelectronic species?

(Given : Atomic Number : $\mathrm{F}: 9, \mathrm{Cl}: 17, \mathrm{Na}=11, \mathrm{Mg}=12, \mathrm{Al}=13, \mathrm{K}=19, \mathrm{Ca}=20, \mathrm{Sc}=21$)

(A) $\mathrm{N}^{3-}, \mathrm{O}^{2-}, \mathrm{F}^{-}, \mathrm{S}^{2-}$

(B) $\mathrm{Ba}^{2+}, \mathrm{Sr}^{2+}, \mathrm{K}^{+}, \mathrm{Ca}^{2+}$

(C) $\mathrm{Li}^{+}, \mathrm{Na}^{+}, \mathrm{Mg}^{2+}, \mathrm{Ca}^{2+}$

(D) $\mathrm{K}^{+}, \mathrm{Cl}^{-}, \mathrm{Ca}^{2+}, \mathrm{Sc}^{3+}$

Open solution

Q113. JEE Main 2023 (Online) 31st January Evening Shift

Arrange the following orbitals in decreasing order of energy.

A. $\mathrm{n}=3, \mathrm{l}=0, \mathrm{m}=0$

B. $\mathrm{n}=4, \mathrm{l}=0, \mathrm{m}=0$

C. $\mathrm{n}=3, \mathrm{l}=1, \mathrm{m}=0$

D. $\mathrm{n}=3, \mathrm{l}=2, \mathrm{m}=1$

The correct option for the order is :

(A) $\mathrm{B}>\mathrm{D}>\mathrm{C}>\mathrm{A}$

(B) $\mathrm{A}>\mathrm{C}>\mathrm{B}>\mathrm{D}$

(C) $\mathrm{D}>\mathrm{B}>\mathrm{A}>\mathrm{C}$

(D) $\mathrm{D}>\mathrm{B}>\mathrm{C}>\mathrm{A}$

Open solution

Q114. JEE Main 2023 (Online) 31st January Morning Shift

Which transition in the hydrogen spectrum would have the same wavelength as the Balmer type transition from $\mathrm{n}=4$ to $\mathrm{n}=2$ of He^{+} spectrum

(A) $\mathrm{n}=3$ to $\mathrm{n}=4$

(B) $\mathrm{n}=2$ to $\mathrm{n}=1$

(C) $\mathrm{n}=1$ to $\mathrm{n}=2$

(D) $\mathrm{n}=1$ to $\mathrm{n}=3$

Open solution

Q115. JEE Main 2023 (Online) 30th January Evening Shift

Maximum number of electrons that can be accommodated in shell with $n=4$ are:

(A) 50

(B) 32

(C) 72

(D) 16

Open solution

Q116. JEE Main 2023 (Online) 30th January Evening Shift

The wave function (Ψ) of $2\,\mathrm{s}$ is given by

$$\Psi_{2\,\mathrm{s}}=\frac{1}{2\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{1/2}\left(2-\frac{r}{a_0}\right)e^{-r/2a_0}$$

At $r=r_0$, radial node is formed. Thus, r_0 in terms of a_0

(A) $r_0=a_0$

(B) $r_0=4a_0$

(C) $r_0=2a_0$

(D) $r_0=\frac{a_0}{2}$

Open solution

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Q117. JEE Main 2023 (Online) 29th January Morning Shift

The shortest wavelength of hydrogen atom in Lyman series is λ. The longest wavelength in Balmer series of He^+ is

(A) $\frac{36\lambda}{5}$ (B) $\frac{5}{9\lambda}$ (C) $\frac{9\lambda}{5}$ (D) $\frac{5\lambda}{9}$

Open solution

Q118. JEE Main 2023 (Online) 25th January Morning Shift

The radius of the 2nd orbit of Li^{2+} is x. The expected radius of the 3rd orbit of Be^{3+} is

(A) $\frac{16}{27}x$ (B) $\frac{4}{9}x$ (C) $\frac{9}{4}x$ (D) $\frac{27}{16}x$

Open solution

Q119. JEE Main 2023 (Online) 24th January Evening Shift

The number of s-electrons present in an ion with 55 protons in its unipositive state is

(A) 10 (B) 9 (C) 12 (D) 8

Open solution

Q120. JEE Main 2023 (Online) 24th January Morning Shift

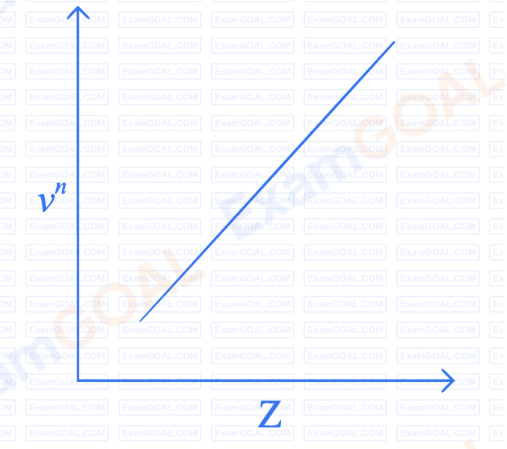
The magnetic moment of a transition metal compound has been calculated to be 3.87 B.M. The metal ion is

(A) Ti^{2+} (B) V^{2+} (C) Cr^{2+} (D) Mn^{2+}

Open solution

Q121. JEE Main 2023 (Online) 24th January Morning Shift

It is observed that characteristic X-ray spectra of elements show regularity. When frequency to the power "n" i.e. v^n of X-rays emitted is plotted against atomic number "Z", following graph is obtained.



The value of 'n' is :

(A) 1 (B) $\frac{1}{2}$ (C) 2 (D) 3

Open solution

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Q122. JEE Main 2022 (Online) 30th July Evening Shift

When the kinetic energy of an electron is increased nine times, the wavelength of the de-Broglie wave associated with it would become :

(A) One-third (B) Three times (C) Two third (D) One sixth

Open solution

Q123. JEE Main 2022 (Online) 29th July Evening Shift

Given below are the quantum numbers for 4 electrons.

A. $\mathrm{n}=3, l=2, \mathrm{m}_l=1, \mathrm{m}_\mathrm{s}=+1/2$

B. $\mathrm{n}=4, l=1, \mathrm{m}_l=0, \mathrm{m}_\mathrm{s}=+1/2$

C. $\mathrm{n}=4, l=2, \mathrm{m}_l=-2, \mathrm{m}_\mathrm{s}=-1/2$

D. $\mathrm{n}=3, l=1, \mathrm{m}_l=-1, \mathrm{m}_\mathrm{s}=+1/2$

The correct order of increasing energy is :

(A) D < B < A < C (B) D < A < B < C (C) B < D < A < C (D) B < D < C < A

Open solution

Q124. JEE Main 2022 (Online) 28th July Evening Shift

Given below are two statements: One is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion $\mathbf{A}$: Zero orbital overlap is an out of phase overlap.

Reason $\mathbf{R}$: It results due to different orientation / direction of approach of orbitals.

In the light of the above statements, choose the correct answer from the options given below

(A) Both A and R are true and R is the correct explanation of A

(B) Both A and R are true but R is NOT the correct explanation of A

(C) A is true but R is false

(D) A is false but R is true

Open solution

Q125. JEE Main 2022 (Online) 28th July Morning Shift

Identify the incorrect statement from the following.

(A) A circular path around the nucleus in which an electron moves is proposed as Bohr's orbit.

(B) An orbital is the one electron wave function (ψ) in an atom.

(C) The existence of Bohr's orbits is supported by hydrogen spectrum.

(D) Atomic orbital is characterised by the quantum numbers n and l only.

Open solution

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Q126. JEE Main 2022 (Online) 27th July Evening Shift

The correct decreasing order of energy for the orbitals having, following set of quantum numbers :

(A) n = 3, l = 0, m = 0

(B) n = 4, l = 0, m = 0

(C) n = 3, l = 1, m = 0

(D) n = 3, l = 2, m = 1

is :

(A) (D) > (B) > (C) > (A)

(B) (B) > (D) > (C) > (A)

(C) (C) > (B) > (D) > (A)

(D) (B) > (C) > (D) > (A)

Open solution

Q127. JEE Main 2022 (Online) 27th July Morning Shift

Given below are two statements. One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Energy of $2s$ orbital of hydrogen atom is greater than that of $2s$ orbital of lithium.

Reason R: Energies of the orbitals in the same subshell decrease with increase in the atomic number.

In the light of the above statements, choose the correct answer from the options given below.

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Open solution

Q128. JEE Main 2022 (Online) 25th July Morning Shift

Which of the following sets of quantum numbers is not allowed?

(A) $n = 3, l = 2, m_l = 0, s = +\frac{1}{2}$

(B) $n = 3, l = 2, m_l = -2, s = +\frac{1}{2}$

(C) $n = 3, l = 3, m_l = -3, s = -\frac{1}{2}$

(D) $n = 3, l = 0, m_l = 0, s = -\frac{1}{2}$

Open solution

Q129. JEE Main 2022 (Online) 30th June Morning Shift

The number of radial nodes and total number of nodes in 4p orbital respectively are :

(A) 2 and 3 **(B)** 2 and 2 **(C)** 3 and 4 **(D)** 4 and 4

Open solution

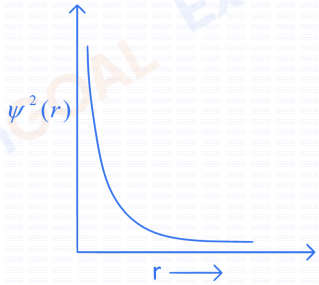
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Q130. JEE Main 2022 (Online) 29th June Evening Shift

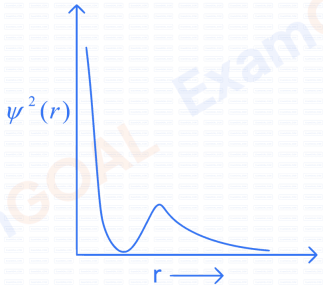
Which of the following is the correct plot for the probability density ψ^2 (r) as a function of distance 'r' of the electron from the nucleus for 2s orbital?

Open solution

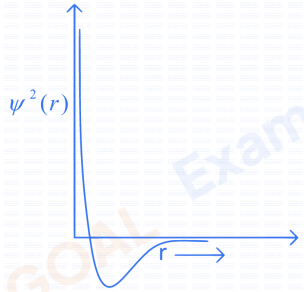
(A)



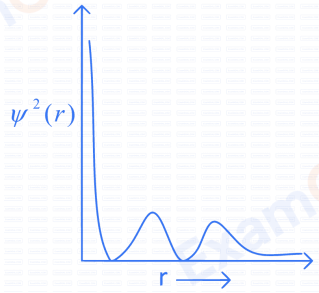
(B)



(C)



(D)



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Q131. JEE Main 2022 (Online) 29th June Morning Shift

Open solution

Which of the following statements are correct?

(A) The electronic configuration of Cr is [Ar] $3d^5\ 4s^1$.

(B) The magnetic quantum number may have a negative value.

(C) In the ground state of an atom, the orbitals are filled in order of their increasing energies.

(D) The total number of nodes are given by n $-$ 2.

Choose the most appropriate answer from the options given below :

(A) (A), (C) and (D) only

(B) (A) and (B) only

(C) (A) and (C) only

(D) (A), (B) and (C) only

Q132. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

Consider the following statements :

(A) The principal quantum number 'n' is a positive integer with values of 'n' = 1, 2, 3, ...

(B) The azimuthal quantum number 'l' for a given 'n' (principal quantum number) can have values as 'l' = 0, 1, 2, ... n

(C) Magnetic orbital quantum number 'm_l' for a particular 'l' (azimuthal quantum number) has (2l + 1) values.

(D) $\pm$ 1/2 are the two possible orientations of electron spin.

(E) For l = 5, there will be a total of 9 orbital

Which of the above statements are correct?

(A) (A), (B) and (C)

(B) (A), (C), (D) and (E)

(C) (A), (C) and (D)

(D) (A), (B), (C) and (D)

Q133. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

The number of radial and angular nodes in 4d orbital are, respectively

(A) 1 and 2

(B) 3 and 2

(C) 1 and 0

(D) 2 and 1

Q134. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

If the radius of the 3rd Bohr's orbit of hydrogen atom is r_3 and the radius of 4th Bohr's orbit is r_4. Then :

(A) $r_4 = \frac{9}{16}r_3$

(B) $r_4 = \frac{16}{9}r_3$

(C) $r_4 = \frac{3}{4}r_3$

(D) $r_4 = \frac{4}{3}r_3$

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Q135. JEE Main 2022 (Online) 25th June Evening Shift

The minimum energy that must be possessed by photons in order to produce the photoelectric effect with platinum metal is :

[Given : The threshold frequency of platinum is $1.3 \times 10^{15}\ s^{-1}$ and $h = 6.6 \times 10^{-34}$ J s.]

(A) 3.21×10^{-14} J (B) 6.24×10^{-16} J (C) 8.58×10^{-19} J (D) 9.76×10^{-20} J

Open solution

Q136. JEE Main 2022 (Online) 25th June Morning Shift

The pair, in which ions are isoelectronic with Al^{3+} is :

(A) Br^- and Be^{2+} (B) Cl^- and Li^+ (C) S^{2-} and K^+ (D) O^{2-} and Mg^{2+}

Open solution

Q137. JEE Main 2022 (Online) 24th June Evening Shift

The energy of one mole of photons of radiation of wavelength 300 nm is

(Given : $h = 6.63 \times 10^{-34}$ J s, $N_A = 6.02 \times 10^{23}\ mol^{-1}$, $c = 3 \times 10^8$ m s^{-1})

(A) 235 kJ mol^{-1} (B) 325 kJ mol^{-1} (C) 399 kJ mol^{-1} (D) 435 kJ mol^{-1}

Open solution

Q138. JEE Main 2022 (Online) 24th June Morning Shift

Consider the following pairs of electrons

(A) (a) n = 3, l = 1, m_l = 1, $m_s = +\frac{1}{2}$

(b) n = 3, l = 2, m_l = 1, $m_s = +\frac{1}{2}$

(B) (a) n = 3, l = 2, $m_l = -2$, $m_s = -\frac{1}{2}$

(b) n = 3, l = 2, $m_l = -1$, $m_s = -\frac{1}{2}$

(C) (a) n = 4, l = 2, m_l = 2, $m_s = +\frac{1}{2}$

(b) n = 3, l = 2, m_l = 2, $m_s = +\frac{1}{2}$

The pairs of electrons present in degenerate orbitals is/are :

(A) Only (A) (B) Only (B) (C) Only (C) (D) (B) and (C)

Open solution

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Q139. JEE Main 2021 (Online) 26th August Morning Shift

Open solution

Given below are two statements.

Statement I : According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in positive charges on the nucleus as there is no strong hold on the electron by the nucleus.

Statement II : According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in principal quantum number.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

Q140. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

Outermost electronic configuration of a group 13 element, E, is $4s^2$, $4p^1$. The electronic configuration of an element of p-block period-five placed diagonally to element, E is :

(A) [Kr] $3d^{10}$ $4s^2$ $4p^2$

(B) [Ar] $3d^{10}$ $4s^2$ $4p^2$

(C) [Xe] $5d^{10}$ $6s^2$ $6p^2$

(D) [Kr] $4d^{10}$ $5s^2$ $5p^2$

Q141. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

Given below are two statements :

Statement I : Bohr's theory accounts for the stability and line spectrum of Li^+ ion.

Statement II : Bohr's theory was unable to explain the splitting of spectral lines in the presence of a magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but statement II is true.

(B) Both statement I and statement II are true.

(C) Statement I is true but statement II is false.

(D) Both statement I and statement II are false.

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Q142. JEE Main 2021 (Online) 18th March Morning Shift

A certain orbital has no angular nodes and two radial nodes. The orbital is :

(A) 2s (B) 3s (C) 3p (D) 2p

Open solution

Q143. JEE Main 2021 (Online) 26th February Evening Shift

Which of the following forms of hydrogen emits low energy β particles?

(A) Tritium ${}_1^3H$ (B) Proton H^+ (C) Protium ${}_1^1H$ (D) Deuterium ${}_1^2H$

Open solution

Q144. JEE Main 2021 (Online) 26th February Morning Shift

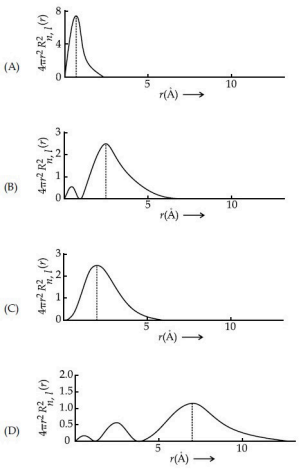
The orbital having two radial as well as two angular nodes is :

(A) 3p (B) 5d (C) 4d (D) 4f

Open solution

Q145. JEE Main 2021 (Online) 25th February Morning Shift

The plots of radial distribution functions for various orbitals of hydrogen atom against 'r' are given below :



The correct plot for 3s orbital is :

(A) (B) (B) (A) (C) (C) (D) (D)

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Q146. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

According to Bohr's atomic theory :

(A) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$.

(B) The product of velocity (v) of electron and principal quantum number (n), $'vn' \propto Z^2$.

(C) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$.

(D) Coulombic force of attraction on the electron is $\propto \frac{Z^3}{n^4}$.

Choose the most appropriate answer from the options given below :

(A) (A), (C) and (D) only
(B) (A) only
(C) (C) only
(D) (A) and (D) only

Q147. JEE Main 2020 (Online) 5th September Evening Slot

Open solution

The correct statement about probability density (except at infinite distance from nucleus) is :

(A) It can be zero for 1s orbital
(B) It can be zero for 3p orbital
(C) It can never be zero for 2s orbital
(D) It can negative for 2p orbital

Q148. JEE Main 2020 (Online) 5th September Morning Slot

Open solution

The difference between the radii of 3rd and 4th orbits of Li^{2+} is ΔR_1 . The difference between the radii of 3rd and 4th orbits of He^+ is ΔR_2 .
Ratio $\Delta R_1 : \Delta R_2$ is :

(A) 8 : 3
(B) 3 : 2
(C) 2 : 3
(D) 3 : 8

Q149. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

The shortest wavelength of H atom in the Lyman series is λ_1. The longest wavelength in the Balmer series of He^+ is :

(A) $\frac{5\lambda_1}{9}$
(B) $\frac{36\lambda_1}{5}$
(C) $\frac{27\lambda_1}{5}$
(D) $\frac{9\lambda_1}{5}$

Q150. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

The region in the electromagnetic spectrum where the Balmar series lines appear is :

(A) Microwave
(B) Ultraviolet
(C) Visible
(D) Infrared

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Q151. JEE Main 2020 (Online) 3rd September Evening Slot

Consider the hypothetical situation where the azimuthal quantum number, l, takes values 0, 1, 2, ..., n + 1, where n is the principal quantum number. Then, the element with atomic number :

(A) 13 has a half-filled valence subshell

(B) 9 is the first alkali metal

(C) 8 is the first noble gas

(D) 6 has a 2p-valence subshell

Open solution

Q152. JEE Main 2020 (Online) 2nd September Evening Slot

The number of subshells associated with n = 4 and m = −2 quantum numbers is

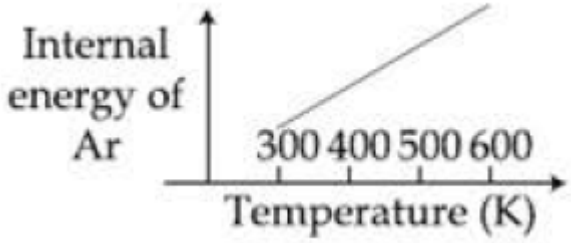
(A) 8 (B) 2 (C) 16 (D) 4

Open solution

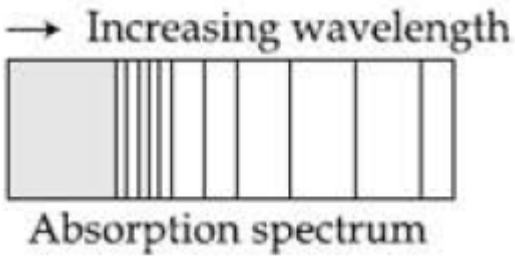
Q153. JEE Main 2020 (Online) 2nd September Morning Slot

The figure that is not a direct manifestation of the quantum nature of atoms is :

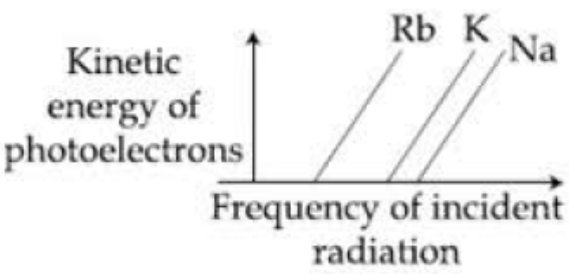
(A)



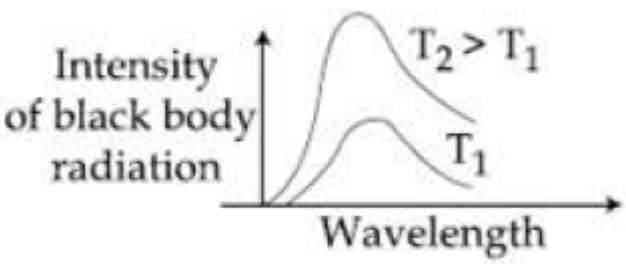
(B)



(C)



(D)



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Q154. JEE Main 2020 (Online) 9th January Morning Slot

The de Broglie wavelength of an electron in the 4th Bohr orbit is :

(A) $2\pi a_0$ (B) $6\pi a_0$ (C) $8\pi a_0$ (D) $4\pi a_0$

Open solution

Q155. JEE Main 2020 (Online) 8th January Evening Slot

The radius of the second Bohr orbit, in terms of the Bohr radius, a_0, in Li^{2+} is :

(A) $\frac{2a_0}{9}$ (B) $\frac{2a_0}{3}$ (C) $\frac{4a_0}{9}$ (D) $\frac{4a_0}{3}$

Open solution

Q156. JEE Main 2020 (Online) 8th January Evening Slot

Hydrogen has three isotopes (A), (B) and (C). If the number of neutron(s) in (A), (B) and (C) respectively, are (x), (y) and (z), the sum of (x), (y) an (z) is :

(A) 3 (B) 1 (C) 4 (D) 2

Open solution

Q157. JEE Main 2020 (Online) 8th January Morning Slot

For the Balmer series in the spectrum of H atom,
$\overline{\nu} = R_H \left\{ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right\}$, the correct statements among (I) to (IV) are :

(I) As wavelength decreases, the lines in the series converge

(II) The integer n_1 is equal to 2

(III) The line of longest wavelength corresponds to $n_2 = 3$

(IV) The ionization energy of hydrogen can be calculated from wave number of these lines

(A) (II), (III), (IV) (B) (I), (II), (III)

(C) (I), (III), (IV) (D) (I), (II), (IV)

Open solution

Q158. JEE Main 2020 (Online) 7th January Morning Slot

The number of orbitals associated with quantum number n = 5, $m_s = +\frac{1}{2}$ is :

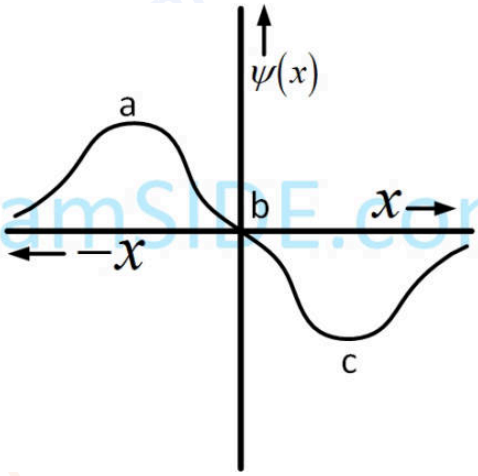
(A) 11 (B) 25 (C) 15 (D) 50

Open solution

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Q159. JEE Main 2019 (Online) 12th April Morning Slot

The electrons are more likely to be found :



Open solution

(A) only in the region a
(B) in the region a and c
(C) only in the region c
(D) in the region a and b

Q160. JEE Main 2019 (Online) 10th April Evening Slot

The ratio of the shortest wavelength of two spectral series of hydrogen spectrum is found to be about 9. The spectral series are :

Open solution

(A) Paschen and Pfund
(B) Balmer and Brackett
(C) Lyman and Paschen
(D) Brackett and Pfund

Q161. JEE Main 2019 (Online) 10th April Morning Slot

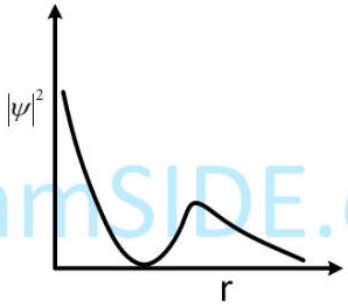
The isoelectronic set of ions is :

Open solution

(A) F^{-}, Li^{+}, Na^{+} and Mg^{2+}
(B) Li^{+}, Na^{+}, O^{2-} and F^{-}
(C) N^{3-}, O^{2-}, F^{-} and Na^{+}
(D) N^{3-}, Li^{+}, Mg^{2+} and O^{2-}

Q162. JEE Main 2019 (Online) 10th April Morning Slot

The graph between $|\psi|^2$ and r (radial distance) is shown below. This represents :



Open solution

(A) 3s orbital
(B) 2s orbital
(C) 2p orbital
(D) 1s orbital

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Q163. JEE Main 2019 (Online) 9th April Evening Slot

Which one of the following about an electron occupying the 1s orbital in a hydrogen atom is incorrect ?

(The Bohr radius is represented by a_0)

(A) The probability density of finding the electron is maximum at the nucleus.

(B) The electron can be found at a distance $2a_0$ from the nucleus

(C) The total energy of the electron is maximum when it is at a distance a_0 from the nucleus.

(D) The magnitude of potential energy is double that of its kinetic energy on an average.

Open solution

Q164. JEE Main 2019 (Online) 9th April Morning Slot

For any given series of spectral lines of atomic hydrogen, let $\Delta\overline{v} = \Delta\overline{v}_{\max} - \Delta\overline{v}_{\min}$ be the difference in maximum and minimum frequencies in cm^{-1}. The ratio Lyman Balmer $\frac{\Delta\overline{v}_{Lyman}}{\Delta\overline{v}_{Balmer}}$ is :

(A) 9 : 4 (B) 4 : 1 (C) 27 : 5 (D) 5 : 4

Open solution

Q165. JEE Main 2019 (Online) 8th April Evening Slot

If p is the momentum of the fastest electron ejected from a metal surface after the irradiation of light having wavelength λ, then for 1.5 p momentum of the photoelectron, the wavelength of the light should be: (Assume kinetic energy of ejected photoelectron to be very high in comparison to work function)

(A) 1/2 λ (B) 3/4 λ (C) 4/9 λ (D) 2/3 λ

Open solution

Q166. JEE Main 2019 (Online) 8th April Morning Slot

The quantum numbers of four electrons are given below :

n = 4, l = 2, m_l =−2, m_s = −1/2

n = 3, l = 2, m_l = 1, m_s = +1/2

n = 4, l = 1, m_l = 0, m_s = +1/2

n = 3, l = 1, m_l = 1, m_s = −1/2

The correct order of their increasing energies will be :

(A) IV < II < III < I (B) I < III < II < IV

(C) IV < III < II < I (D) I < II < III < IV

Open solution

Q167. JEE Main 2019 (Online) 12th January Evening Slot

If the de Broglie wavelength of the electron in n^{th} Bohr orbit in a hydrogenic atom is equal to 1.5 πa_0 (a_0 is Bohr radius), then the value of n/z is -

(A) 0.75 (B) 0.40 (C) 1.50 (D) 1.0

Open solution

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Q168. JEE Main 2019 (Online) 12th January Morning Slot

Open solution

What is the work function of the metal if the light of wavelength 4000$\mathring{A}$ generates photoelectrons of velocity 6×10^5 ms^{-1} from it ?
(Mass of electron = 9×10^{-31} kg;
Velocity of light = 3×10^8 ms^{-1}
Plank's constant = 6.626×10^{-34} Js;
Charge of electron = 1.6×10^{-19} JeV^{-1})

(A) 4.0 eV (B) 0.9 eV (C) 2.1 eV (D) 3.1 eV

Q169. JEE Main 2019 (Online) 11th January Evening Slot

Open solution

The de Broglie wavelength (λ) associated with a photoelectron varies with the frequency (v) of the incident radiation as, [v_0 is threshold frequency] :

(A) $\lambda \propto \frac{1}{(v-v_0)^{\frac{3}{2}}}$ (B) $\lambda \propto \frac{1}{(v-v_0)^{\frac{1}{4}}}$

(C) $\lambda \propto \frac{1}{(v-v_0)}$ (D) $\lambda \propto \frac{1}{(v-v_0)^{\frac{1}{2}}}$

Q170. JEE Main 2019 (Online) 11th January Morning Slot

Open solution

Heat treatment of muscular pain involves radiation of wavelength of about 900 nm. Which spectral line of H atom is suitable for this purpose?

[$R_H = 1 \times 10^5$ cm^{-1}, $h = 6.6 \times 10^{-34}$ Js, $c = 3 \times 10^8$ ms^{-1}]

(A) Balmer, $\infty \rightarrow 2$ (B) Paschen, $5 \rightarrow 3$

(C) Paschen, $\infty \rightarrow 3$ (D) Lyman, $\infty \rightarrow 1$

Q171. JEE Main 2019 (Online) 10th January Evening Slot

Open solution

The ground state energy of hydrogen atom is – 13.6 eV. The energy of second excited state of He^+ ion in eV is :

(A) $-$ 6.04 (B) $-$ 54.4 (C) $-$ 27.2 (D) $-$ 3.4

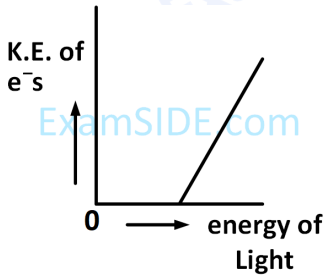
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Q172. JEE Main 2019 (Online) 10th January Morning Slot

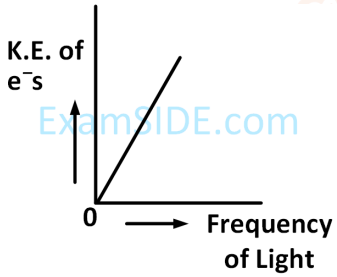
Which of the graphs shown below does not represent the relationship between incident light and the electron ejected from metal surface ?

Open solution

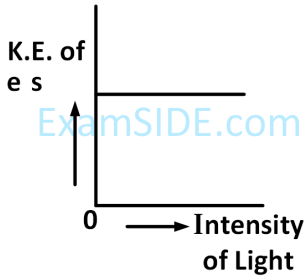
(A)



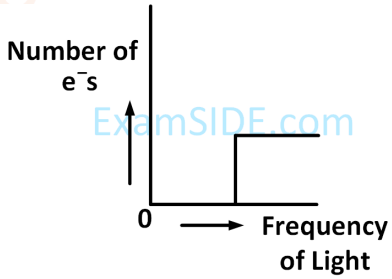
(B)



(C)



(D)



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Q173. JEE Main 2019 (Online) 9th January Evening Slot

Which of the following combination of statements is true regarding the interpretation of the atomic orbitals ?

Open solution

(a) An electron in an orbital of high angular momentum stays away from the nucleus than an electron in the orbital of lower angular momentum.

(b) For a given value of the principal quantum number, the size of the orbit is inversely proportional to the azimuthal quantum number.

(c) According to wave mechanics, the ground state angular momentum is equal to $\frac{h}{2\pi}$.

(d) The plot of ψ vs r for various azimuthal quantum numbers, shows peak shifting towards higher r value.

(A) (a), (b) **(B)** (a), (d) **(C)** (b), (c) **(D)** (a), (c)

Q174. JEE Main 2019 (Online) 9th January Morning Slot

For emission line of atomic hydrogen from n_i = 8 to n_f = n, the plot of wave number $(\overline{v})$ against $\left(\frac{1}{n^2}\right)$ will be (The Rydberg constant, R_H is in wave number unit)

Open solution

(A) Linear with intercept $-R_H$ **(B)** Non linear

(C) Linear with slope R_H **(D)** Linear with slope $-R_H$

Q175. JEE Main 2018 (Online) 16th April Morning Slot

Which of the following statements is **false** ?

Open solution

(A) Photon has momentum as well as wavelength.

(B) Splitting of spectral lines in electrical field is called Stark effect.

(C) Rydberg constant has unit of energy.

(D) Frequency of emitted radiation from a black body goes from a lower wavelength to higher wavelength as the temperature increases.

Q176. JEE Main 2018 (Online) 15th April Evening Slot

The de-Broglie's wavelength of electron present in first Bohr orbit of 'H' atom is :

Open solution

(A) $0.529\ \mathring{A}$ **(B)** $2\pi \times 0.529\ \mathring{A}$

(C) $\frac{0.529}{2\pi}\ \mathring{A}$ **(D)** $4 \times 0.529\ \mathring{A}$

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Q177. JEE Main 2018 (Online) 15th April Morning Slot

Ejection of the photoelectron from metal in the photoelectric experiment can be stopped by applying 0.5 V when the radiation of 250 nm is used. The work function of the metal is :

(A) 4 eV (B) 4.5 eV (C) 5 eV (D) 5.5 eV

Open solution

Q178. JEE Main 2017 (Online) 9th April Morning Slot

The electron in the hydrogen atom undergoes transition from higher orbitals to orbital of radius 211.6 pm. This transition is associated with :

(A) Lyman series (B) Balmer series (C) Paschen series (D) Brackett series

Open solution

Q179. JEE Main 2017 (Online) 8th April Morning Slot

If the shortest wavelength in Lyman series of hydrogen atom is A, then the longest wavelength in Paschen series of He^+ is :

(A) $\frac{5A}{9}$ (B) $\frac{9A}{5}$ (C) $\frac{36A}{5}$ (D) $\frac{36A}{7}$

Open solution

Q180. JEE Main 2017 (Offline)

The radius of the second Bohr orbit for hydrogen atom is:

(Planck's Const. h = 6.6262 × 10^{-34} Js; mass of electron = 9.1091 × 10^{-31} kg; charge of electron (e) = 1.60210 × 10^{-19} C; permittivity of vacuum (ε_0) = 8.854185 × 10^{-12} kg^{-1} m^{-3} A^2)

(A) 4.76 $\overset{o}{A}$ (B) 2.12 $\overset{o}{A}$

(C) 0.529 $\overset{o}{A}$ (D) 1.65 $\overset{o}{A}$

Open solution

Q181. JEE Main 2017 (Offline)

The group having isoelectronic species is:

(A) O^{2-} , F^- , Na^+, Mg^{2+} (B) O^{2-} , F^- , Na, Mg^{2+}

(C) O^- , F^- , Na^+, Mg^{2+} (D) O^- , F^- , Na, Mg^+

Open solution

Q182. JEE Main 2016 (Online) 10th April Morning Slot

Aqueous solution of which salt will not contain ions with the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6$?

(A) NaF (B) NaCl (C) KBr (D) CaI_2

Open solution

Q183. JEE Main 2016 (Online) 9th April Morning Slot

The total number of orbitals associated with the principal quantum number 5 is :

(A) 5 (B) 10 (C) 20 (D) 25

Open solution

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Q184. JEE Main 2016 (Offline)

A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference V esu. If e and m are charge and mass of an electron, respectively, then the value of h / λ (where λ is wavelength associated with electron wave) is given by :

(A) 2meV (B) $\sqrt{meV}$ (C) $\sqrt{2meV}$ (D) meV

Open solution

Q185. JEE Main 2015 (Online) 11th April Morning Slot

At temperature T, the average kinetic energy of any particle is $\frac{3}{2}$ KT. The de Broglie wavelength follows the order :

(A) Thermal proton > Visible photon > Thermal electron

(B) Thermal electron > Thermal proton > Visible photon

(C) Visible photon > Thermal electron > Thermal neutron

(D) Visible photon > Thermal neutron > Thermal electron

Open solution

Q186. JEE Main 2015 (Online) 10th April Morning Slot

If the principal quantum number n = 6, the correct sequence of filling of electrons will be :

(A) ns $\rightarrow$np$\rightarrow$(n $-$ 1)d$\rightarrow$ (n $-$ 2)f (B) ns $\rightarrow$ (n $-$ 2)f $\rightarrow$ (n $-$ 1)d $\rightarrow$ np

(C) ns $\rightarrow$ (n $-$ 1)d $\rightarrow$ (n $-$ 2)f $\rightarrow$ np (D) ns $\rightarrow$ (n $-$ 2)f $\rightarrow$ np $\rightarrow$(n $-$ 1) d

Open solution

Q187. JEE Main 2015 (Offline)

Which of the following is the energy of a possible excited state of hydrogen?

(A) −6.8 eV (B) −3.4 eV (C) +6.8 eV (D) +13.6 eV

Open solution

Q188. JEE Main 2014 (Online) 19th April Morning Slot

Ionization energy of gaseous Na atoms is 495.5 kJ mol^{-1}. The lowest possible frequency of light that ionizes a sodium atom is

(h = 6.626 $\times$ 10^{-34} Js, N_A = 6.022 $\times$ 10^{23} mol^{-1})

(A) 7.50 $\times$ 10^{4} s^{-1} (B) 4.76 $\times$ 10^{14} s^{-1}

(C) 3.15 $\times$ 10^{15} s^{-1} (D) 1.24 $\times$ 10^{15} s^{-1}

Open solution

Q189. JEE Main 2014 (Online) 12th April Morning Slot

If m and e are the mass and charge of the revolving electron in the orbit of radius r for hydrogen atom, the total energy of the revolving electron will be :

(A) $\frac{1}{2}\frac{e^2}{r}$ (B) $-\frac{e^2}{r}$

(C) $\frac{me^2}{r}$ (D) $-\frac{1}{2}\frac{e^2}{r}$

Open solution

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Q190. JEE Main 2014 (Online) 12th April Morning Slot

The de-Broglie wavelength of a particle of mass 6.63 g moving with a velocity of 100 ms^{-1} is :

(A) 10^{-33} m (B) 10^{-35} m (C) 10^{-31} m (D) 10^{-25} m

Open solution

Q191. JEE Main 2014 (Online) 12th April Morning Slot

Excited hydrogen atom emits light in the ultraviolet region at 2.47×10^{15} Hz. With this frequency, the energy of a single photon is :
($h = 6.63 \times 10^{-34}$ Js)

(A) 8.041×10^{-40} J (B) 2.680×10^{-19} J
(C) 1.640×10^{-18} J (D) 6.111×10^{-17} J

Open solution

Q192. JEE Main 2014 (Online) 11th April Morning Slot

If λ_0 and λ be the threshold wavelength and wavelength of incident light, the velocity of photoelectron ejected from the metal surface is :

(A) $\sqrt{\frac{2h}{m}(\lambda_0 - \lambda)}$
(B) $\sqrt{\frac{2hc}{m}(\lambda_0 - \lambda)}$
(C) $\sqrt{\frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)}$
(D) $\sqrt{\frac{2h}{m}\left(\frac{1}{\lambda_0} - \frac{1}{\lambda}\right)}$

Open solution

Q193. JEE Main 2014 (Online) 11th April Morning Slot

Based on the equation :

$\Delta E = -2.0 \times 10^{-18}\ J \left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$

the wavelength of the light that must be absorbed to excite hydrogen electron from level n = 1 to level n = 2 will be :
($h = 6.625 \times 10^{-34}$ Js, $C = 3 \times 10^8$ ms^{-1})

(A) 1.325×10^{-7} m (B) 1.325×10^{-10} m
(C) 2.650×10^{-7} m (D) 5.300×10^{-10} m

Open solution

Q194. JEE Main 2014 (Online) 9th April Morning Slot

The energy of an electron in first Bohr orbit of H - atom is -13.6 eV. The energy value of electron in the excited state of Li^{2+} is :

(A) -27.2 eV (B) 30.6 eV (C) -30.6 eV (D) 27.2 eV

Open solution

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Q195. JEE Main 2014 (Offline)

The correct set of four quantum numbers for the valence electrons of rubidium atom (Z= 37) is:

(A) 5, 1, 1, + 1/2 (B) 5, 1, 0, + 1/2 (C) 5, 0, 0, + 1/2 (D) 5, 0, 1, + 1/2

Open solution

Q196. JEE Main 2013 (Online) 25th April Morning Slot

Given (A) n = 5, m_l = +1 (B) n = 2, l = 1, m_l = -1, m_s = -1/2. The maximum number of electron(s) in an atom that can have the quantum numbers as given in (A) and (B) are respectively:

(A) 25 and 1 (B) 8 and 1 (C) 2 and 4 (D) 4 and 1

Open solution

Q197. JEE Main 2013 (Online) 23th April Morning Slot

The de Broglie wavelength of a car of mass 1000 kg and velocity 36 km/hr is :

(A) 6.626×10^{-34} m (B) 6.626×10^{-38} m (C) 6.626×10^{-31} m (D) 6.626×10^{-30} m

Open solution

Q198. JEE Main 2013 (Online) 22th April Morning Slot

The wave number of the first emission line in the Balmer series of H-Spectrum is : (R = Rydberg constant)

(A) $\frac{5}{36}R$ (B) $\frac{9}{400}R$ (C) $\frac{7}{6}R$ (D) $\frac{3}{4}R$

Open solution

Q199. JEE Main 2013 (Online) 9th April Morning Slot

In an atom how many orbital(s) will have the quantum numbers; $n = 3$, $l = 2$ and $m_l = +2$?

(A) 5 (B) 3 (C) 1 (D) 7

Open solution

Q200. JEE Main 2013 (Offline)

Energy of an electron is given by $E = -2.178 \times 10^{-18} J \left(\frac{Z^2}{n^2}\right)$. Wavelength of light required to excite an electron in an hydrogen atom from level n = 1 to n = 2 will be
(h = 6.62 × 10^{-34} Js and c = 3.0 × 10^{8} ms^{-1})

(A) 2.816 × 10^{-7} m (B) 6.500 × 10^{-7} m (C) 8.500 × 10^{-7} m (D) 1.214 × 10^{-7} m

Open solution

Q201. AIEEE 2012 (Online) 26th May

The following sets of quantum numbers represent four electrons in an atom. (i) $n = 4, l = 1$ (ii) $n = 4, l = 0$ (iii) $n = 3, l = 2$ (iv) $n = 3, l = 1$ The sequence representing increasing order of energy, is

(A) (iii) < (i) < (iv) < (ii) (B) (iv) < (ii) < (iii) < (i)
(C) (i) < (iii) < (ii) < (iv) (D) (ii) < (iv) < (i) < (iii)

Open solution

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Q202. AIEEE 2012 (Online) 19th May

If the kinetic energy of an electron is increased four times, the wavelength of the de-Broglie wave associated with it would become

(A) one fourth (B) half (C) four times (D) two times

Open solution

Q203. AIEEE 2012 (Online) 12th May

If the radius of first orbit of H atom is a_0, the de Broglie wavelength of an electron in the third orbit is

(A) $4\pi a_0$ (B) $8\pi a_0$ (C) $6\pi a_0$ (D) $2\pi a_0$

Open solution

Q204. AIEEE 2012 (Online) 7th May

The limiting line in Balmer series will have a frequency of (Rydberg constant, $R_\infty = 3.29 \times 10^{15}$ cycles/s)

(A) $8.22 \times 10^{14}\ \text{s}^{-1}$ (B) $3.29 \times 10^{15}\ \text{s}^{-1}$

(C) $3.65 \times 10^{14}\ \text{s}^{-1}$ (D) $5.26 \times 10^{13}\ \text{s}^{-1}$

Open solution

Q205. AIEEE 2012

The electrons identified by quantum numbers n and l :

(a) n = 4, l = 1

(b) n = 4, l = 0

(c) n = 3, l = 2

(d) n = 3, l = 1

Can be placed in order of increasing energy as :

(A) (c) < (d) < (b) < (a) (B) (d) < (b) < (c) < (a)

(C) (b) < (d) < (a) < (c) (D) (a) < (c) < (b) < (d)

Open solution

Q206. AIEEE 2011

A gas absorbs a photon of 355 nm and emits at two wavelengths. If one of the emissions is at 680 nm, the other is at :

(A) 325 nm (B) 743 nm (C) 518 nm (D) 1035 nm

Open solution

Q207. AIEEE 2010

The energy required to break one mole of Cl–Cl bonds in Cl_2 is 242 kJ mol^{-1}. The longest wavelength of light capable of breaking a single Cl – Cl bond is

(c = 3 x 10^8 ms^{-1} and N_A = 6.02 x 10^{23} mol^{-1})

(A) 594 nm (B) 640 nm (C) 700 nm (D) 494 nm

Open solution

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Q208. AIEEE 2010

Ionisation energy of He^+ is 19.6 x 10^{-18} J $atom^{-1}$. The energy of the first stationary state (n = 1) of Li^{2+} is

(A) 4.41 x 10^{-16} J $atom^{-1}$
(B) -4.41 x 10^{-17} J $atom^{-1}$
(C) -2.2 x 10^{-15} J $atom^{-1}$
(D) 8.82 x 10^{-17} J $atom^{-1}$

Open solution

Q209. AIEEE 2009

Calculate the wavelength (in nanometer) associated with a proton moving at 1.0 x 10^3 ms^{-1} (Mass of proton = 1.67 $\times$ 10^{-27} kg and h = 6.63 $\times$ 10^{-34} Js) :

(A) 0.40 nm
(B) 2.5 nm
(C) 14.0 nm
(D) 0.032 nm

Open solution

Q210. AIEEE 2009

In an atom, an electron is moving with a speed of 600 m/s with an accuracy of 0.005%. Certainity with which the position of the electron can be located is (h = 6.6 $\times$ 10^{-34} kg m^2s^{-1}, mass of electron, e_m = 9.1 $\times$ 10^{-31} kg)

(A) 5.10 $\times$ 10^{-3} m
(B) 1.92 $\times$ 10^{-3} m
(C) 3.84 $\times$ 10^{-3} m
(D) 1.52 $\times$ 10^{-4} m

Open solution

Q211. AIEEE 2008

Which one of the following constitutes a group of the isoelectronic species?

(A) C_2^{2-}, O_2^-, CO, NO
(B) NO^+, C_2^{2-}, CN^-, N_2
(C) CN^-, N_2, O_2^{2-}, C_2^{2-}
(D) N_2, O_2^-, NO^+, CO

Open solution

Q212. AIEEE 2008

The ionization enthalpy of hydrogen atom is 1.312 × 10^6 J mol^{-1}. The energy required to excite the electron in the atom from n = 1 to n = 2 is

(A) 8.51 × 10^5 J mol^{-1}
(B) 6.56 × 10^5 J mol^{-1}
(C) 7.56 × 10^5 J mol^{-1}
(D) 9.84 × 10^5 J mol^{-1}

Open solution

Q213. AIEEE 2007

Which of the following sets of quantum numbers represents the highest energy of an atom?

(A) n = 3, l = 0, m = 0, s = +1/ 2
(B) n = 3, l = 1, m = 1, s = +1/ 2
(C) n = 3, l = 2, m = 1, s = +1/ 2
(D) n = 4, l = 0, m = 0, s = +1/ 2

Open solution

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Q214. AIEEE 2006

According to Bohr's theory, the angular momentum of an electron in 5th orbit is

(A) 10 h/π (B) 2.5 h/π (C) 25 h/π (D) 1.0 h/π

Open solution

Q215. AIEEE 2006

Uncertainty in the position of an electron (mass = 9.1 $\times$ 10^{-31} kg) moving with a velocity 300 ms^{-1}, accurate up to 0.001% will be (h = 6.63 $\times$ 10^{-34} Js)

(A) 1.92 $\times$ 10^{-2} m (B) 3.84 $\times$ 10^{-2} m (C) 19.2 $\times$ 10^{-2} m (D) 5.76 $\times$ 10^{-2} m

Open solution

Q216. AIEEE 2006

Which of the following sets of ions represents a collection of isoelectronic species?

(A) N^{3-}, O^{2-}, F^-, S^{2-} (B) Li^+, Na^+, Mg^{2+}, Ca^{2+}

(C) K^+, Cl^-, Ca^{2+}, Sc^{3+} (D) Ba^{2+}, Sr^{2+}, K^+, Ca^{2+}

Open solution

Q217. AIEEE 2005

In a multi-electron atom, which of the following orbitals described by the three quantum numbers will have the same energy in the absence of magnetic and electric fields?

(A) n = 1, l = 0, m = 0
(B) n = 2, l = 0, m = 0
(C) n = 2, l = 1, m = 1
(D) n = 3, l = 2, m = 1
(E) n = 3, l = 2, m = 0

(A) (D) and (E) (B) (C) and (D) (C) (B) and (C) (D) (A) and (B)

Open solution

Q218. AIEEE 2005

Of the following sets which one does NOT contain isoelectronic species?

(A) $BO_3^{3-}, CO_3^{2-}, NO_3^-$ (B) $SO_3^{2-}, CO_3^{2-}, NO_3^-$

(C) CN^-, N_2, C_2^{2-} (D) $PO_4^{3-}, SO_4^{2-}, ClO_4^-$

Open solution

Q219. AIEEE 2005

Pick out the isoelectronic structure from the following :

(i) CH_3^+
(ii) H_3O^+
(iii) NH_3
(iv) CH_3^-

(A) (i) and (ii) (B) (iii) and (iv) (C) (i) and (iii) (D) (ii), (iii) and (iv)

Open solution

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Q220. AIEEE 2004

Which of the following sets of quantum numbers is correct for an electron in 4f orbital?

(A) n = 4, l = 3, m = +1, s = + 1/2
(B) n = 4, l = 4, m = -4, s = - 1/2
(C) n = 4, l = 3, m = +4, s = + 1/2
(D) n = 3, l = 2, m = -2, s = + 1/2

Open solution

Q221. AIEEE 2004

Consider the ground state of Cr atom (Z = 24). The number of electrons with the azimuthal quantum numbers, l = 1 and 2 are respectively

(A) 16 and 4
(B) 12 and 5
(C) 12 and 4
(D) 16 and 5

Open solution

Q222. AIEEE 2004

The wavelength of the radiation emitted when in a hydrogen atom electron falls from infinity to stationary state 1, would be (Rydberg constant = 1.097×10^7 m^{-1})

(A) 406 nm
(B) 192 nm
(C) 91 nm
(D) 9.1×10^{-8} nm

Open solution

Q223. AIEEE 2004

Which one of the following sets of ions represents the collection of isoelectronic species? (Atomic nos. : F = 9, Cl = 17, Na = 11, Mg = 12, Al = 13, K = 19, Ca = 20, Sc = 21)

(A) K^+, Cl^-, Mg^{2+}, Sc^{3+}
(B) Na^+, Ca^{2+}, Sc^{3+}, F^-
(C) K^+, Ca^{2+}, Sc^{3+}, Cl^-
(D) Na^+, Mg^{2+}, Al^{3+}, Cl^-

Open solution

Q224. AIEEE 2003

The orbital angular momentum for an electron revolving in an orbit is given by $\sqrt{l(l+1)}\frac{h}{2\pi}$. This momentum for an s-electron will be given by

(A) zero
(B) $\frac{h}{2\pi}$
(C) $\sqrt{2}\frac{h}{2\pi}$
(D) $+\frac{1}{2}\frac{h}{2\pi}$

Open solution

Q225. AIEEE 2003

Which one of the following groupings represents a collection of isoelectronic species? (At. nos. : Cs : 55, Br : 35)

(A) N^{3-}, F^-, Na^+
(B) Be, Al^{3+}, Cl^-
(C) Ca^{2+}, Cs^+, Br
(D) Na^+, Ca^{2+}, Mg^{2+}

Open solution

Q226. AIEEE 2003

In Balmer series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of hydrogen

(A) $5 \rightarrow 2$
(B) $4 \rightarrow 1$
(C) $2 \rightarrow 5$
(D) $3 \rightarrow 2$

Open solution

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Q227. AIEEE 2003

The de Broglie wavelength of a tennis ball of mass 60 g moving with a velocity of 10 meters per second is approximately

(A) 10^{-31} meters (B) 10^{-16} meters (C) 10^{-25} meters (D) 10^{-33} meters

Open solution

Q228. AIEEE 2002

In a hydrogen atom, if energy of an electron in ground state is -13.6 eV, then that in the 2nd excited state is

(A) -1.51 eV (B) -3.4 eV (C) -6.04 eV (D) -13.6 eV

Open solution

Q229. AIEEE 2002

Uncertainty in position of a minute particle of mass 25 g in space is 10^{-5} m. What is the uncertainty in its velocity (in ms^{-1}) ($h = 6.6 \times 10^{-34}$ Js)

(A) 2.4×10^{-34} (B) 0.5×10^{-34} (C) 2.1×10^{-28} (D) 0.5×10^{-23}

Open solution

Out of syllabus 4 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q230. JEE Main 2023 (Online) 8th April Morning Shift · **Numerical**

The number of following statement/s which is/are incorrect is _____

(A) Line emission spectra are used to study the electronic structure

(B) The emission spectra of atoms in the gas phase show a continuous spread of wavelength from red to violet

(C) An absorption spectrum is like the photographic negative of an emission spectrum

(D) The element helium was discovered in the sun by spectroscopic method

Open solution

Q231. JEE Main 2021 (Online) 27th July Evening Shift · **MCQ (Single Correct Answer)**

If the Thomson model of the atom was correct, then the result of Rutherford's gold foil experiment would have been :

(A) All of the α-particles pass through the gold foil without decrease in speed.

(B) α-particles are deflected over a wide range of angles.

(C) All α-particles get bounced back by 180°

(D) α-particles pass through the gold foil deflected by small angles and with reduced speed.

Open solution

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Q232. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Rutherford's gold foil experiment cannot explain the line spectrum of hydrogen atom.

Statement II : Bohr's model of hydrogen atom contradicts Heisenberg's uncertainty principle.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is false but statement II is true.
(B) Statement I is true but statement II is false.
(C) Both statement I and statement II are false.
(D) Both statement I and statement II are true.

Q233. AIEEE 2005 · MCQ (Single Correct Answer)

Open solution

Which of the following statements in relation to the hydrogen atom is correct?

(A) 3s orbital is lower in energy than 3p orbital
(B) 3p orbital is lower in energy than 3d orbital
(C) 3s and 3p orbitals are of lower energy than 3d orbital
(D) 3s, 3p and 3d orbitals all have the same energy

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Answer Key

Numerical

1. **3**	2. **5**	3. **2**	4. **54**	5. **54**
6. **58**	7. **1724**	8. **5**	9. **34**	10. **658**
11. **6**	12. **22**	13. **4**	14. **8**	15. **494**
16. **10**	17. **16**	18. **5**	19. **0**	20. **3**
21. **3**	22. **1**	23. **0**	24. **3**	25. **7**
26. **2**	27. **798**	28. **270**	29. **5**	30. **492**
31. **548**	32. **2**	33. **4**	34. **1758**	35. **10**
36. **300**	37. **2**	38. **22**	39. **4**	40. **2**
41. **7**	42. **50**	43. **3155**	44. **5**	45. **58**
46. **2**	47. **6**	48. **0**	49. **0**	50. **3**
51. **9**	52. **1**	53. **181**	54. **1**	55. **2**
56. **222**				

MCQ (Single Correct Answer)

57. **D**	58. **A**	59. **D**	60. **B**	61. **D**
62. **B**	63. **B**	64. **B**	65. **D**	66. **C**
67. **C**	68. **A**	69. **B**	70. **A**	71. **C**
72. **B**	73. **D**	74. **D**	75. **B**	76. **D**
77. **C**	78. **A**	79. **C**	80. **B**	81. **C**
82. **B**	83. **B**	84. **B**	85. **A**	86. **A**
87. **C**	88. **D**	89. **B**	90. **A**	91. **D**
92. **D**	93. **C**	94. **A**	95. **D**	96. **D**
97. **A**	98. **B**	99. **C**	100. **A**	101. **C**
102. **D**	103. **B**	104. **A**	105. **A**	106. **B**
107. **C**	108. **C**	109. **C**	110. **B**	111. **C**
112. **D**	113. **D**	114. **B**	115. **B**	116. **C**
117. **C**	118. **D**	119. **A**	120. **B**	121. **B**
122. **A**	123. **B**	124. **A**	125. **D**	126. **A**
127. **A**	128. **C**	129. **A**	130. **B**	131. **D**
132. **C**	133. **A**	134. **B**	135. **C**	136. **D**
137. **C**	138. **B**	139. **C**	140. **D**	141. **A**

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142. **B**	143. **A**	144. **B**	145. **D**	146. **D**
147. **B**	148. **C**	149. **D**	150. **C**	151. **A**
152. **B**	153. **A**	154. **C**	155. **D**	156. **A**
157. **B**	158. **B**	159. **B**	160. **C**	161. **C**
162. **B**	163. **C**	164. **A**	165. **C**	166. **A**
167. **A**	168. **C**	169. **D**	170. **C**	171. **A**
172. **B**	173. **B**	174. **C**	175. **D**	176. **B**
177. **B**	178. **B**	179. **D**	180. **B**	181. **A**
182. **A**	183. **D**	184. **C**	185. **C**	186. **B**
187. **B**	188. **D**	189. **D**	190. **A**	191. **C**
192. **C**	193. **A**	194. **C**	195. **C**	196. **B**
197. **B**	198. **A**	199. **C**	200. **D**	201. **B**
202. **B**	203. **C**	204. **A**	205. **B**	206. **B**
207. **D**	208. **B**	209. **A**	210. **B**	211. **B**
212. **D**	213. **C**	214. **B**	215. **A**	216. **C**
217. **A**	218. **B**	219. **D**	220. **A**	221. **B**
222. **C**	223. **C**	224. **A**	225. **A**	226. **A**
227. **D**	228. **A**	229. **C**		

Out of syllabus

230. **1**	231. **D**	232. **D**	233. **D**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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